

Entropy and Mutual Information in Dijet Hadron Multiplicities: Motivation, Interpretation Limits, and a Finite-Statistics Methodology

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Abstract

The mutual information (MI) between the hadron multiplicities of back-to-back jets is a new, information-theoretically complete measure of inter-jet correlation, motivated by the entanglement-entropy interpretation of hadron multiplicity distributions [1, 3, 4, 5]. We first establish precisely what this observable can and cannot probe. Because hadron counting is a single, fixed measurement, the joint multiplicity distribution is a classical probability distribution, and its Shannon conditional entropy and normalized MI obey $S(B|A) \geq 0$ and $\text{NMI} \leq 1$ *identically* – these bounds cannot be violated by any dataset, whether or not the underlying partonic state is entangled, so multiplicity-based entropies cannot serve as a quantum witness. Their physics value lies elsewhere: ansatz-conditional tests of the maximal-entanglement picture of hadronization, an information-complete generalization of forward–backward multiplicity correlations, discrimination between hadronization models, and a quantitative classical-correlation baseline for the broader quantum-information program at colliders. The core of this note is the statistical problem of measuring these entropies reliably at finite statistics. We show that the naive plug-in entropy/MI estimator is systematically biased upward at finite event count N , in exactly the direction that mimics a genuine correlation excess. Using a controlled Monte Carlo toy model with a known ground truth, we validate two standard corrections (Miller–Madow and shuffle-subtraction) and a more rigorous Bayesian estimator (NSB, Nemenman–Shafee–Bialek). We then validate the full pipeline – cross-checked between independent Python and ROOT/C++ implementations – on realistic PYTHIA 8 dijet Monte Carlo, introduce the scale-comparable normalized observable $\text{NMI} = I(A:B)/\min(S_A, S_B)$, and document a genuine bootstrap-consistency pitfall uncovered during cross-validation together with its fix. Finally, since PYTHIA 8 output has no independently known true MI, we develop a Tier-2 closure test that calibrates a synthetic model to the real data’s own statistics to obtain a trustworthy pseudo-truth. Applied to the whole validation sample, this shows NSB’s bootstrap uncertainty under-covers by a factor of roughly 1.5–1.7; applied differentially in leading-jet- p_T bins, with a separate calibrated model fit per bin, the same test finds substantially better-calibrated (though still imperfect) coverage, suggesting the whole-sample under-coverage is at least partly an artifact of fitting one model across a range of p_T -dependent structure. Both results are reported as quantitative calibration factors to be applied explicitly before quoting any uncertainty on real detector data, not as corrections silently folded into the pipeline.

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1 Physics motivation and what this observable can teach

Take a back-to-back dijet event (e.g. from e^+e^- , DIS, or pp collisions at RHIC/STAR). Treat jet A and jet B as the two subsystems of a bipartite system whose “state” is accessed only through observed hadron multiplicities, via **local parton–hadron duality (LPHD)**: the hadron multiplicity distribution stands in for the underlying parton Fock-state distribution.

The Schmidt-decomposition ansatz [1, 2]: if each Fock/parton-number sector n carries probability $P(n)$, and the reduced density matrix of a jet is diagonal in n , then the entanglement entropy of that jet reduces to the **Shannon entropy of its hadron multiplicity distribution**,

$$S_A = - \sum_n P_A(n) \ln P_A(n), \quad (1)$$

and likewise for S_B . This identification underlies the successful comparison of measured hadron entropies with entanglement-entropy predictions in DIS [3, 4] and, recently, in jet fragmentation [5].

The measurement proposed here: by additionally recording the *joint* multiplicity distribution $P_{AB}(n_A, n_B)$ across both jets in the same event, one can construct the mutual information $I(A:B)$, the conditional entropy $S(B|A)$, and a normalized mutual information NMI (all defined in Sec. 2) – extending the single-jet entropy program to the two-jet, correlation level.

An interpretation limit, stated up front. As shown in detail in Sec. 2.1, these multiplicity-based entropies *cannot* be used to witness or certify quantum correlations: the measured joint distribution is, by construction, a classical probability distribution, for which $S(B|A) \geq 0$ and $\text{NMI} \leq 1$ hold as mathematical identities – no data, however entangled its partonic origin, can violate them. The motivation for the measurement is therefore *not* a classical-bound-violation test. (Sec. 2.3 shows how models with genuine quantum correlations are nonetheless confronted with the data: by projecting the model down to the measured number-basis statistics, never by lifting the data up to the state.) What the observable genuinely teaches is the following.

1. **Ansatz-conditional tests of the entanglement picture of hadronization.** Under the Schmidt/LPHD ansatz, a *pure* bipartite dijet state predicts equal von Neumann marginal entropies, hence (via the ansatz) the testable relation $S_A \simeq S_B$ even for kinematically asymmetric jet selections; and in the idealized Schmidt limit the two multiplicities are perfectly correlated, $\text{NMI} \rightarrow 1$. Where the measured NMI sits between the independent limit ($\text{NMI} = 0$) and the Schmidt limit ($\text{NMI} = 1$) quantifies how much of the initial state’s correlation structure survives hadronization as classical correlation – extending the single-jet entropy tests of Refs. [3, 4, 5] to the joint two-jet level.
2. **An information-complete generalization of forward–backward multiplicity correlations.** The classic forward–backward correlation strength b_{corr} [15] is a linear (second-moment) statistic. $I(A:B)$ captures *all* statistical dependence between the two multiplicities at once – every moment, including nonlinear structure – and NMI makes the result dimensionless and comparable across collision energies, kinematic selections, and experiments. No dijet multiplicity MI measurement exists to date; it is a new correlation observable even as pure QCD.
3. **Hadronization-model discrimination.** String and cluster hadronization predict different cross-jet correlation topologies – in e^+e^- dijets a single string spans both hemispheres, so string dynamics predicts specific long-range hemisphere multiplicity correlations that cluster models do not share. Differential MI vs. jet p_T and cone radius R is a tuning-grade generator constraint that b_{corr} alone does not provide.
4. **Separation of classical correlation sources.** In pp collisions, cross-jet correlations mix conservation laws (energy–momentum, charge, strangeness – calculable), color coherence (concentrated at small R), initial-state fluctuations, and multi-parton interactions / underlying event (scaling with event activity). MI measured differentially in p_T , R , and event activity decomposes these contributions.

5. **Sensitivity to (though not certification of) quantum dynamics.** Quantum interference alters the *diagonal* $P_{AB}(n_A, n_B)$ even after the coherences are lost: coherent vs. independent gluon emission, Bose enhancement, and color reconnection all leave imprints in the joint multiplicity distribution. A measured MI in excess of classical-cascade generator baselines would be evidence of dynamics those generators lack – a discovery-grade anomaly, even though not a nonclassicality certificate (Sec. 2.1).
6. **A rigorous classical-correlation envelope.** For any underlying quantum state, the number-basis Shannon MI measured here is a lower bound on the classical-correlation content $J(A:B)$ of that state. It is therefore a required input to any future decomposition of total vs. classical correlations using other degrees of freedom (e.g. spin), a program outside the scope of this note.
7. **Extensions to nuclear environments.** Dijet multiplicity MI in AA relative to pp measures medium-induced decorrelation of the jet pair – an entropy-language counterpart of jet quenching – and at the EIC, current-jet vs. target-fragmentation entropy correlations probe the x -dependence of the initial state’s entanglement structure, the original context of Ref. [1].

Why this is subtle in practice: every one of the measurements above requires entropies of a two-dimensional multiplicity histogram estimated from a finite event sample, and MI estimated from a finite sample is **always statistically biased** – as shown below, biased in exactly the direction that mimics a genuine correlation excess if not handled carefully. The central technical content of this note is therefore statistical: establishing, with controlled ground truth and realistic generator validation, which estimators can be trusted, at what sample sizes, with what calibrated uncertainties.

2 The observables, their classical bounds, and why they cannot witness quantum correlations

From the per-jet and joint hadron-multiplicity distributions we can build three entropies and two derived quantities:

$$S_A, \quad S_B, \quad S_{AB} \quad \implies \quad I(A:B) = S_A + S_B - S_{AB}, \quad S(B|A) = S_{AB} - S_A. \quad (2)$$

Elementary bounds. For *any* joint probability distribution, two classical inequalities hold [6]: *subadditivity*, $S_{AB} \leq S_A + S_B$, equivalent to $I(A:B) \geq 0$; and *monotonicity*, $S_{AB} \geq \max(S_A, S_B)$ (“conditioning cannot make entropy negative”), equivalent to

$$S(B|A) \geq 0, \quad S(A|B) \geq 0, \quad \iff \quad I(A:B) \leq \min(S_A, S_B). \quad (3)$$

The one-line proof of monotonicity: $S(B|A) = \sum_a p(a) S(B|A=a)$ is a convex combination of ordinary entropies, each of which is non-negative. (We are careful to call Eq. (3) monotonicity rather than subadditivity: the distinction matters below, because quantum von Neumann entropy *obeys* subadditivity but *violates* monotonicity for entangled states [7].)

A scale-comparable form: normalized mutual information. Absolute $I(A:B)$ is awkward to compare across different collision energies, kinematic selections, or p_T bins, since S_A and S_B

themselves grow with e.g. jet p_T (Sec. 8.2) and so set a different natural scale for I in every bin. The $\min(S_A, S_B)$ appearing in Eq. (3) already does the needed normalization:

$$\text{NMI} \equiv \frac{I(A:B)}{\min(S_A, S_B)}. \quad (4)$$

By monotonicity, $\text{NMI} \in [0, 1]$ for any distribution: $\text{NMI} = 0$ is the independent limit and $\text{NMI} = 1$ is perfect correlation – the Schmidt limit of the entanglement ansatz (Sec. 1). Since NMI is dimensionless and bounded, it is directly comparable across p_T bins, run periods, or even experiments in a way raw $I(A:B)$ is not. This is a standard normalization in the wider MI literature (one of several conventional choices, e.g. Kvålseth [8]; min-normalization is the tightest of these).

2.1 Why no multiplicity measurement can violate these bounds

It is tempting to hope that, since the quantum (von Neumann) conditional entropy of an entangled state can be negative [7], a sufficiently precise measurement of $S(B|A)$ from multiplicities could reveal $S(B|A) < 0$ (equivalently $\text{NMI} > 1$) and thereby certify a quantum signature. This hope fails – not for practical reasons, but in principle. The argument has four steps, in increasing order of physical content.

(i) The bounds are properties of distributions, not of physics. Any dataset of event-by-event pairs (n_A, n_B) defines a joint probability distribution – the histogram itself. Monotonicity, Eq. (3), holds for *every* joint probability distribution, with no assumption about the dynamics that generated it. There is therefore no possible measured P_{AB} – from PYTHIA, from STAR data, or from the decay of a maximally entangled partonic state – whose population-level $S(B|A)$ is negative or whose NMI exceeds one. The bound is arithmetic, not a null hypothesis.

(ii) Where quantum negativity lives, and why counting cannot reach it. Negative conditional entropy is a property of the von Neumann entropies of the quantum state ρ_{AB} *before* measurement. Writing ρ_{AB} in the number basis, the recorded counts determine exactly the diagonal matrix elements, $\langle n_A, n_B | \rho_{AB} | n_A, n_B \rangle = P_{AB}(n_A, n_B)$, and nothing else: every off-diagonal element (coherence between different multiplicity sectors) drops out of the counting statistics identically, for any number of events. But the von Neumann entropy is a function of the *eigenvalues* of the full matrix, and the eigenvalues depend on the coherences. A minimal example makes the information loss exact: the entangled pure state $|\psi\rangle = \sum_n \sqrt{\lambda_n} |n\rangle_A |n\rangle_B$ (with $S(\rho_{AB}) = 0$) and the classical mixture $\rho = \sum_n \lambda_n |n, n\rangle \langle n, n|$ (with $S(\rho_{AB}) = -\sum_n \lambda_n \ln \lambda_n > 0$) produce *identical* joint counts, $P_{AB}(n_A, n_B) = \lambda_{n_A} \delta_{n_A n_B}$, event after event. Two states whose joint entropies differ by several nats generate exactly the same dataset, so $S(\rho_{AB})$ is simply not a function of the data recorded. Measuring N_A and N_B *simultaneously* does not help: the two are a *commuting* pair – a single measurement setting – just as measuring σ_z on arbitrarily many copies of a qubit pins down $\langle \sigma_z \rangle$ perfectly while leaving $\langle \sigma_x \rangle$, $\langle \sigma_y \rangle$, and hence the state and its entropy, undetermined. More fundamentally, entropy is not an observable in the quantum-mechanical sense: it is nonlinear in ρ , so no Hermitian operator has $S(\rho)$ as its expectation value. Every laboratory determination of an entropy proceeds either through full state tomography (requiring measurements in non-commuting bases) or through collective measurements on multiple copies (e.g. a swap test yielding $\text{Tr } \rho^2$, which requires coherently interfering two events with each other – something no detector does; a large event sample is many separate destructive measurements, not one joint coherent one). Equivalently, in entropy language: measurement in the number basis dephases ρ_{AB} to its diagonal, and dephasing never decreases entropy, so the Shannon S_{AB} of the counts is an *upper bound* on the von

Neumann $S(\rho_{AB})$, with the gap being exactly the coherence content – the excess entropy added by dephasing destroys precisely the negativity one hoped to observe.

(iii) No complementary basis exists for multiplicity. One might try to repair (ii) by measuring in a second, non-commuting basis – the strategy underlying every Bell-type test. For hadron multiplicity no such basis exists, even in principle. Final-state hadron number is a superselected, classically recorded quantity: no detector observable connects different-multiplicity sectors coherently. More broadly, *every* function of the final-state hadron momenta – multiplicity, p_T -weighted multiplicity, jet shapes, substructure observables, alternative cone radii – commutes with every other such function, since all are functions of the same recorded phase-space configuration. Enlarging the set of jet observables therefore only enlarges one classical joint distribution; it never supplies the basis rotation a witness requires. Certifying nonclassical correlations at a collider requires a degree of freedom with an analyzable post-hoc basis choice – for example hadron spin accessed through self-analyzing weak decays [10, 11] – which is a separate program, outside the scope of this note.

(iv) The estimator-level corollary. The plug-in estimate obeys $\widehat{S(B|A)} \geq 0$ *exactly*, for every finite dataset, because the empirical distribution is itself a probability distribution to which Eq. (3) applies. Only bias-corrected estimators (Miller–Madow, NSB; Secs. 5–6) can return $\widehat{S(B|A)} < 0$ or $\widehat{\text{NMI}} > 1$, and only through estimation error. Any apparently significant violation of the classical bounds in an analysis is therefore *guaranteed* to be a finite-sample artifact, and must shrink as N grows. This reframes the statistical task of this note: the estimator machinery below is not protecting a potential discovery claim from false positives – it ensures that the measured classical correlation quantities are unbiased and carry honest uncertainties, and that boundary-hugging artifacts ($\widehat{S(B|A)}$ near or below zero, $\widehat{\text{NMI}}$ near or above one) are recognized as estimator pathology rather than physics.

2.2 What the entanglement ansatz does and does not license

The identification of the marginal Shannon entropy S_A with an entanglement entropy is a *model statement*: it requires the Schmidt/LPHD assumptions (a near-pure bipartite state whose reduced density matrices are diagonal in the number sectors). Under those assumptions the measurement acquires interpretive power: $S_A \simeq S_B$ tests the purity prediction; the measured entropies can be compared against entanglement-entropy predictions computed from parton distributions or fragmentation functions [1, 3, 4, 5]; and the position of NMI within $[0, 1]$ measures how close the jet pair sits to the Schmidt (perfectly correlated) limit. These are consistency tests of an entanglement-based model against alternatives – legitimate and falsifiable – but consistency cannot be promoted to certification: a classical mechanism producing the same joint distribution is never excluded by this observable alone (Sec. 2.1).

Quantum discord is likewise out of reach. True (Ollivier–Zurek [9]) quantum discord is defined by optimizing conditional entropy over quantum measurement bases – non-commuting projective measurements that do not reduce to reading off a classical value. For any classical joint distribution – and by (ii) above, hadron-multiplicity data is one – the optimal “measurement” on B is trivially observing N_B exactly, which makes the classical-correlation piece $J(A:B)$ equal to the *full* $I(A:B)$, so discord $D = I - J \equiv 0$ identically, with no exception. Extracting a genuinely nonzero discord requires actual quantum-state tomography of a different degree of freedom [11, 10],

not multiplicity counting. What the multiplicity measurement contributes to that broader program is the classical-correlation envelope of Sec. 1, item 6 – the rest of this note is about measuring it reliably.

2.3 Comparing to models with quantum correlations

The no-go of Sec. 2.1 does not prevent confronting the data with a model in which the dijet system is genuinely entangled; it only fixes the *direction* of the comparison. The data can never be lifted up to $S(\rho_{AB})$, but a quantum model can always be projected down to the data: by the Born rule, the model predicts the outcome statistics of the measurement actually performed. The comparison therefore happens at the Shannon level, where the data lives, with the model’s von Neumann quantities acting as internal parameters whose *fingerprints* on the number-basis diagonal are what is tested.

The construction has three layers.

1. **The state.** Take the entanglement-based model of the dijet in Schmidt form,

$$|\psi\rangle = \sum_n \sqrt{\lambda_n} |n\rangle_A |n\rangle_B, \quad (5)$$

with the Schmidt weights λ_n supplied by QCD – from small- x evolution / $xG(x, Q^2)$ in the Kharzeev–Levin picture [1], or from fragmentation functions as in Ref. [5]. The model’s quantum-information content is explicit and computable: $S(\rho_{AB}) = 0$ (purity), $S(\rho_A) = S(\rho_B) = -\sum_n \lambda_n \ln \lambda_n$ (the entanglement entropy), and a negative quantum conditional entropy, $S(B|A)_{\text{vN}} = -S(\rho_A) < 0$. None of this is directly measurable – it is the model’s internal bookkeeping.

2. **Dephasing (the measurement).** The predicted parton-level joint distribution is the number-basis diagonal, $P(n, m) = \lambda_n \delta_{nm}$. Note what happens to each quantum quantity under this projection. The marginal entropy survives intact: ρ_A is diagonal in n , so its von Neumann entropy *equals* the Shannon entropy of the marginal counts – this is exactly what the Schmidt/LPHD ansatz buys, and why the single-jet entropy program [3, 4, 5] is well posed. The joint purity does *not* survive: the Shannon entropy of the diagonal is S_A , not zero. The unobservable negativity is replaced by its measurable shadow, **perfect correlation**: $S(B|A) = 0$ and $\text{NMI} = 1$ at parton level.
3. **Hadronization as a channel.** Hadrons are not partons, so each jet’s multiplicity is smeared by a local transfer kernel, $T(N_A|n)$ and $T'(N_B|n)$ (LPHD in its sharpest form is the identity kernel; more realistically a Poisson- or NBD-like smearing around n). The full prediction is

$$P_{AB}(N_A, N_B) = \sum_n \lambda_n T(N_A|n) T'(N_B|n). \quad (6)$$

From Eq. (6) the model’s S_A , S_B , $I(A:B)$, NMI , and $S(B|A)$ follow – all Shannon quantities, directly comparable to the measured ones with the estimator and calibration machinery of the rest of this note.

Where the quantum content shows up in measurables. The model’s quantumness translates into three concrete, falsifiable predictions on the diagonal: purity of the state predicts $S_A \simeq S_B$ even for kinematically asymmetric selections; the entanglement-entropy identification predicts the

value of S_A from xG or the fragmentation functions; and the Schmidt structure predicts NMI near its maximum, degraded from 1 only by the calculable smearing of the T kernels – whereas classical cascade baselines sit at $\text{NMI} \ll 1$ (Sec. 8) with no reason to approach the ceiling. The measurement therefore discriminates the entanglement-structured prediction from generator baselines quantitatively. It is a genuine model comparison – it is just not a certification.

The honest limit: the degeneracy class. Equation (6) depends on ρ_{AB} only through its diagonal. The pure state $|\psi\rangle$ and its fully dephased classical twin, $\rho = \sum_n \lambda_n |n, n\rangle\langle n, n|$ – with $S(\rho_{AB}) = 0$ and $-\sum_n \lambda_n \ln \lambda_n$, respectively – predict *identical* data. The measurement thus constrains the *diagonal equivalence class* of states; within that class, the value of $S(\rho_{AB})$ is a theoretical commitment, not an observable. This is the no-go of Sec. 2.1 seen from the model side, and it should be stated in any comparison: “the data agree with the entanglement-based prediction” and “the data agree with its dephased classical twin” are the same sentence. Distinguishing members of the class requires access to the coherences – a different degree of freedom, e.g. spin [10, 11], outside the scope of this note.

Connection to the toy model. Equation (6) is a latent variable (the Schmidt index n , drawn from λ_n) feeding two conditionally independent emission kernels. This is structurally identical to the Gamma-Poisson toy model of Sec. 4: the latent Λ plays the role of a continuous Schmidt spectrum, and the Poissons play the role of T . The toy model used throughout this note is, in other words, the dephased image of a Schmidt-form state – so the estimator, bias, and closure machinery developed below applies to quantum-model predictions without modification, and the eventual data comparison reduces to a sharp, well-posed question: which latent spectrum and kernels does nature use – those predicted by QCD entanglement (λ_n from xG or fragmentation functions), or those produced by a classical cascade?

3 The statistical problem: finite-sample bias

Every quantity above is estimated from a finite event sample, and the naive (“plug-in”) entropy estimator has a **systematic, structural bias** that turns out to point in exactly the dangerous direction for this measurement.

Where the bias comes from. The naive estimator replaces true probabilities p_i with empirical frequencies $\hat{p}_i = n_i/N$,

$$\hat{S} = - \sum_i \hat{p}_i \ln \hat{p}_i. \quad (7)$$

Because $x \ln x$ is convex (equivalently $-x \ln x$ is concave), Jensen’s inequality guarantees

$$\langle \hat{S} \rangle < S_{\text{true}} \quad \text{for any finite } N, \quad (8)$$

i.e. the plug-in estimator always underestimates entropy. This isn’t a numerical artifact; with only N draws one can never fully resolve a distribution’s tails, and that missing resolution always looks like *less* apparent randomness than there truly is.

Leading-order size of the effect (Miller 1955 [12]), for a distribution over K possible outcomes:

$$\langle \hat{S} \rangle \approx S_{\text{true}} - \frac{K-1}{2N}. \quad (9)$$

Fewer events (N small) or finer binning (K large) both make the downward bias worse.

Why this flips sign and biases $I(A:B)$ upward. Combining the individual biases of $\hat{S}_A, \hat{S}_B, \hat{S}_{AB}$ with correct signs,

$$\langle \hat{I} \rangle \approx I_{\text{true}} + \frac{(K_{AB} - 1) - (K_A - 1) - (K_B - 1)}{2N} = I_{\text{true}} + \frac{K_{AB} - K_A - K_B + 1}{2N}. \quad (10)$$

The joint distribution lives in a much larger space than either marginal alone, so $\hat{S}_{AB}$ suffers a *much larger* downward bias than $\hat{S}_A + \hat{S}_B$ does – and since S_{AB} enters with a **minus** sign, its extra negative bias flips into a **positive** bias on $\hat{I}$. In the extreme case $N \ll K_A K_B$ (badly undersampled joint histogram), this produces large *phantom* apparent mutual information even for truly independent variables – sampling noise alone can masquerade as a real correlation excess. By the same mechanism it pushes $\widehat{S(B|A)}$ spuriously downward toward – and, for bias-corrected estimators, even past – the classical boundary that Sec. 2.1 showed the true quantity can never cross: finite statistics manufactures exactly the boundary-hugging artifacts that make uncorrected entropy measurements untrustworthy.

A commonly-quoted shortcut version of Eq. (10) is $(K_A - 1)(K_B - 1)/(2N)$; this is not a different formula, it is Eq. (10) evaluated under the extra assumption $K_{AB} = K_A K_B$, i.e. a fully and uniformly occupied joint space. That assumption only holds for independent variables; for correlated data the observed K_{AB} is much smaller than $K_A K_B$ (the joint distribution concentrates on a narrow ridge rather than filling the full grid), so this shortcut badly *overestimates* the bias – a pitfall we hit directly in Sec. 5.

Practical consequence. Comparing the same underlying physics across data samples of different size – different STAR running periods, different luminosity, different jet-cone or multiplicity binning choices – will report systematically different central MI (and $S(B|A)$) values purely from this effect, with zero change in the actual QCD physics. Secs. 4–6 demonstrate this concretely with a toy model and work through three ways to correct for it; Secs. 8–9 then validate the corrected pipeline on realistic generator output and calibrate its uncertainties.

4 Toy Monte Carlo test bed

To demonstrate the bias concretely and validate corrections against a known ground truth, we use a controlled toy model: two back-to-back jets, A and B , with a shared latent hard-process fluctuation Λ (e.g. fluctuating parton virtuality / initial-state energy scale) drawn from a Gamma distribution. Conditional on Λ , each jet’s hadron multiplicity is Poisson distributed,

$$\Lambda \sim \text{Gamma}(k, \theta), \quad N_A | \Lambda \sim \text{Poisson}(\Lambda), \quad N_B | \Lambda \sim \text{Poisson}(c\Lambda). \quad (11)$$

Marginalizing over Λ gives each jet a **Negative Binomial** multiplicity distribution (the standard KNO-like form used for hadron multiplicities), and the shared Λ induces a genuine – but purely **classical** – correlation between N_A and N_B , i.e. $I_{\text{true}} > 0$ with $S(B|A)_{\text{true}}$ solidly positive. This is a convenient sandbox: we know the true joint distribution (to arbitrary precision, from a huge reference sample), so we can measure the *bias* of any estimator directly. Parameters used throughout: $k = 3.0$, $\theta = 8.0$, $c = 0.9$ (giving $\text{corr}(N_A, N_B) \approx 0.88$). As noted in Sec. 2.3, this shared-latent structure is also exactly the dephased image of a Schmidt-form entangled state, with Λ playing the role of a continuous Schmidt spectrum – the machinery validated here therefore applies unchanged to the predictions of entanglement-based models.

4.1 Demonstration: two samples, same true distribution, different size

We generate a “small” event sample ($N = 500$; e.g. a single limited run / low-luminosity STAR-like dataset), a “large” event sample ($N = 200,000$; high-statistics dataset), and a very large reference sample ($N = 5 \times 10^6$) to pin down I_{true} – all drawn from the *exact same underlying physics*.¹

Table 1: Naive $\hat{I}$ for three samples of the same true physics.

Sample	N	K_A	K_B	K_{AB}	$\hat{I}_{\text{naive}}$ (nats)
small	500	67	60	390	1.7424
large	200,000	129	113	3949	0.7463
reference	5,000,000	159	147	7022	0.7329 ($\approx I_{\text{true}}$)

We see $\hat{I}_{\text{small}} > \hat{I}_{\text{large}} \approx \hat{I}_{\text{ref}} \equiv I_{\text{true}}$ – **the same physics, analyzed with fewer events, reports a systematically larger MI**. This is the bias predicted in Sec. 3, not a change in physics.

The bias comes entirely from how well-populated the joint histogram is. Figure 1 shows (top) the marginal hadron-multiplicity distributions $P_A(n)$, overlaid for the three samples – these look reasonably similar even at $N = 500$ because there are only $O(10^2)$ bins to fill – and (bottom) the 2D joint histograms $P_{AB}(n_A, n_B)$ on the same color scale, where the difference is dramatic: at $N = 500$ the joint plane is almost empty, while at $N = 5 \times 10^6$ it is a smooth, fully-resolved correlated ridge. A sparse, “salt-and-pepper” joint histogram is exactly what drives $\hat{S}_{AB}$ artificially low, and hence $\hat{I}$ artificially high.

4.2 Bias vs. sample size

Repeating the naive estimate over a range of N (with several repetitions per N to see the scatter) and plotting $\hat{I}_{\text{naive}}$ against $1/N$ (Fig. 2) shows the systematic downward drift (bias shrinking) as N grows, converging toward the flat reference line, and the expected near-linear behavior in $1/N$: the intercept of that line is a cheap, model-independent estimate of the bias-corrected/true MI. The extrapolated intercept, 0.9427 nats, overshoots the reference value of 0.7329 nats by roughly 30%: the bias is only asymptotically linear in $1/N$, because the number of occupied joint bins K_{AB} itself grows with N and so the effective slope in Eq. (10) changes across the fitted range. The extrapolation is a useful cheap diagnostic, not a substitute for the estimators that follow.

5 Corrections: Miller–Madow and shuffle-subtraction

5.1 Miller–Madow correction

The exact leading-order Miller–Madow correction [12] applies the $-(K-1)/(2N)$ term to *each* of S_A, S_B, S_{AB} separately and combines them:

$$\hat{I}_{MM} = \hat{I}_{\text{naive}} - \frac{(K_{AB} - 1) - (K_A - 1) - (K_B - 1)}{2N} = \hat{I}_{\text{naive}} - \frac{K_{AB} - K_A - K_B + 1}{2N}. \quad (12)$$

Crucially, K_{AB} here must be the *actually observed* number of occupied joint bins, not the product $K_A K_B$: for correlated variables (our case, by construction) the joint distribution concentrates on

¹The toy-model tables in this and the following sections (Tables 1, 2, and 3) were each generated from independent random draws; entries at the same nominal N therefore differ between tables by ordinary sampling scatter. That scatter is itself a preview of the estimator variance quantified in Secs. 5–6.

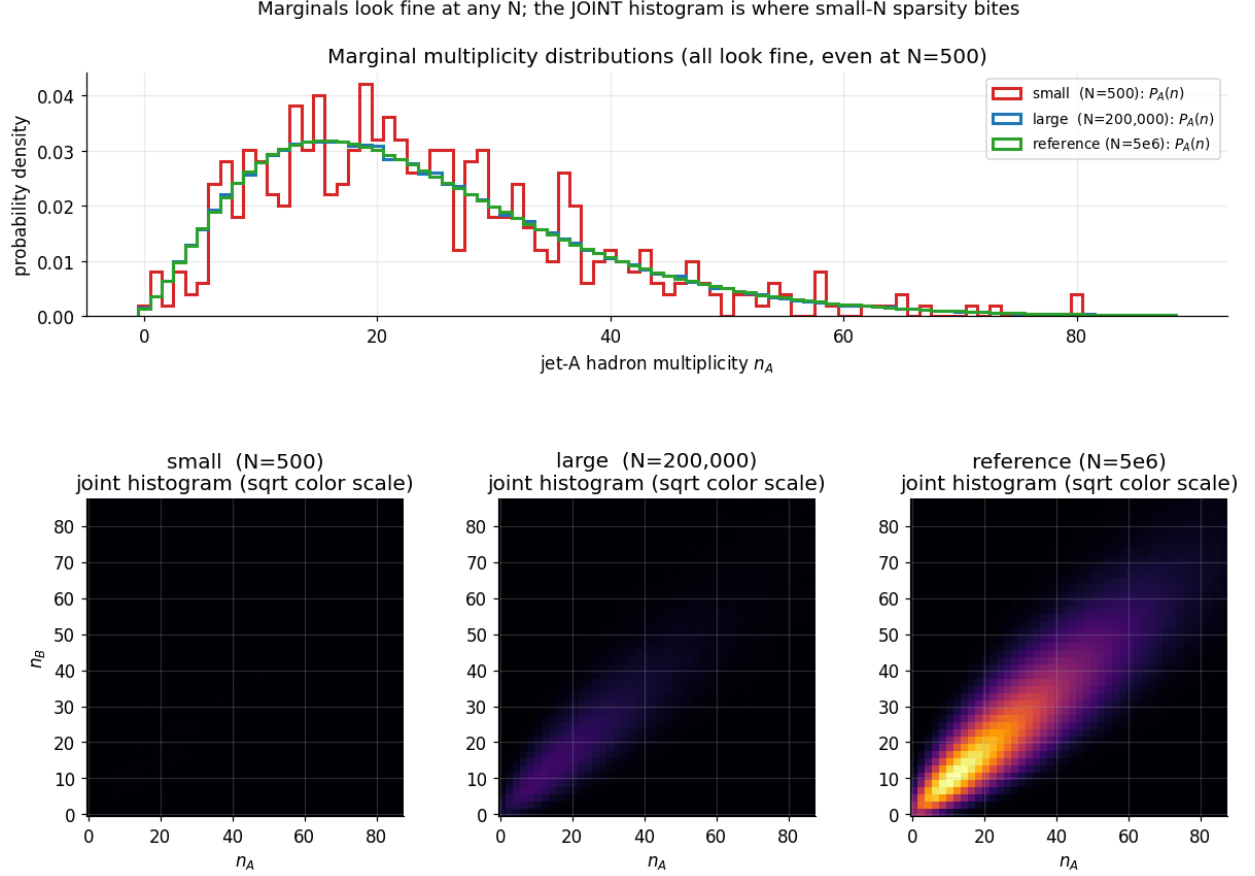


Figure 1: Marginal multiplicity distributions (top) and 2D joint histograms (bottom, square-root color scale) for the small, large, and reference samples. Marginals look consistent at any N ; the joint histogram is where small- N sparsity bites.

a narrow ridge and $K_{AB} \ll K_A K_B$, so using the product in place of the observed value wildly overcorrects and can drive $\hat{I}_{MM}$ negative.

5.2 Shuffle-subtraction (label-permutation null)

Randomly permuting the B -jet labels relative to the A -jet labels *within the same sample* destroys any true correlation ($I_{\text{true}, \text{shuffled}} = 0$ by construction) while reproducing the same finite-sample bias. Averaging over many shuffles gives a direct empirical estimate of the bias to subtract:

$$\hat{I}_{\text{corrected}} = \hat{I}_{\text{naive}} - \langle \hat{I}_{\text{shuffled}} \rangle. \quad (13)$$

This makes no asymptotic assumption and remains reliable even when $N \lesssim K_A K_B$, where Miller–Madow starts to break down. It has its own bias, however: shuffling spreads the joint distribution over *more* bins than the true correlated data occupies, so the shuffled surrogate is *more* under-sampled than the real data at the same N . Shuffle-subtraction therefore tends to systematically **overestimate** the bias and **overcorrect**, pulling $\hat{I}_{\text{corrected}}$ below I_{true} when the true correlation is strong.

With the correct Miller–Madow formula (using the observed K_{AB}), both corrections pull the small- N estimates much closer to I_{true} than the naive value and stay well-behaved (no more diving

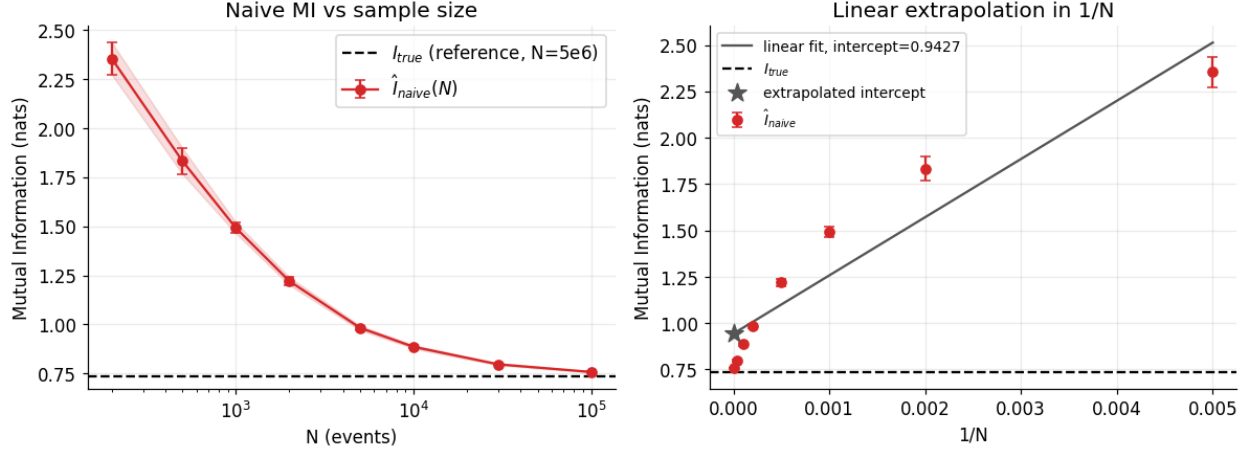


Figure 2: Left: naive $\hat{I}$ vs. sample size N , converging to the reference value. Right: the same data plotted against $1/N$, with a linear extrapolation to $1/N \rightarrow 0$.

Table 2: Naive, Miller–Madow, and shuffle-corrected $\hat{I}$ vs. N .

N	$\hat{I}_{\text{naive}}$	$\hat{I}_{MM}$	$\hat{I}_{\text{shuffle}}$	I_{true} (ref.)
200	2.2764	2.0889	0.0657	0.7329
500	1.8694	1.6014	0.1436	0.7329
1000	1.4271	1.1756	0.2422	0.7329
2000	1.1873	0.9866	0.3809	0.7329
5000	0.9766	0.8539	0.5592	0.7329
20000	0.8259	0.7750	0.6792	0.7329

negative). Neither lands exactly on I_{true} at very small N : Miller–Madow is only a first-order truncation of the true bias expansion, and shuffle-subtraction overcorrects when the true correlation is strong, as explained above. Both effects shrink as N grows.

The shuffle procedure gives more than a bias correction – the spread of $\hat{I}_{\text{shuffled}}$ over many permutations *is* the null distribution for “ $I_{\text{true}} = 0$ ” at that N and binning (Fig. 3). Plotting $\hat{I}_{\text{naive}}$ against that null distribution is a direct visual significance check: at $N = 500$ the null distribution itself is already centered well above zero (pure bias) and the naive value sits close to or inside this null cloud, so on its own it would not be significant evidence of correlation, even though the jets are truly correlated by construction. By $N = 20,000$ the null has collapsed toward zero and the naive value clearly separates from it.

Figure 4 summarizes naive vs. corrected MI across sample sizes, and Fig. 5 plots $|\hat{I} - I_{\text{true}}|$ on a log-log scale, which is the honest way to compare convergence rates. Both corrections sit visibly below the naive curve at fixed N ; where the corrected residual curves run roughly parallel to the naive curve rather than dropping faster, that indicates the residual error is increasingly dominated by sub-leading bias terms that neither method captures (Miller–Madow: higher-order $1/N^2, \dots$ terms; shuffle: the systematic overcorrection discussed above) – exactly the gap the more rigorous fix in Sec. 6 is designed to close.

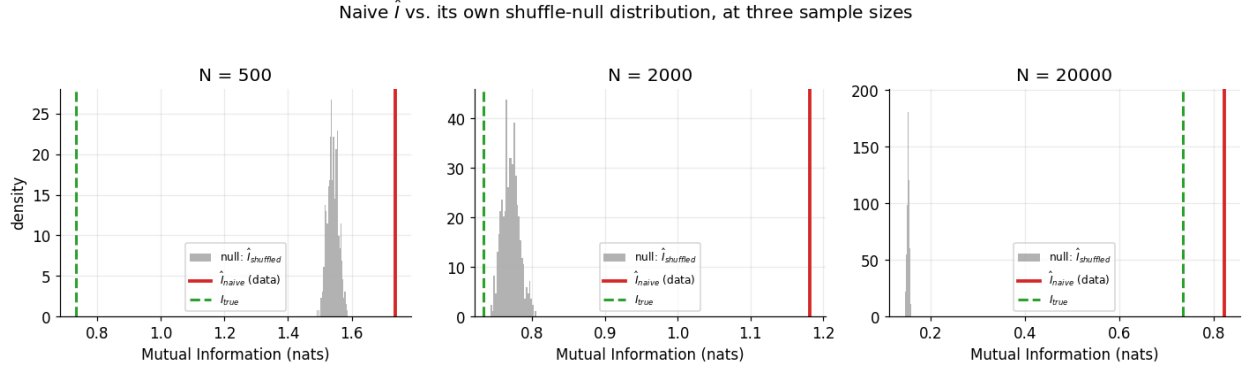


Figure 3: Naive $\hat{I}$ (vertical line) against its own shuffle-null distribution (histogram), at three sample sizes. Dashed line: true value.

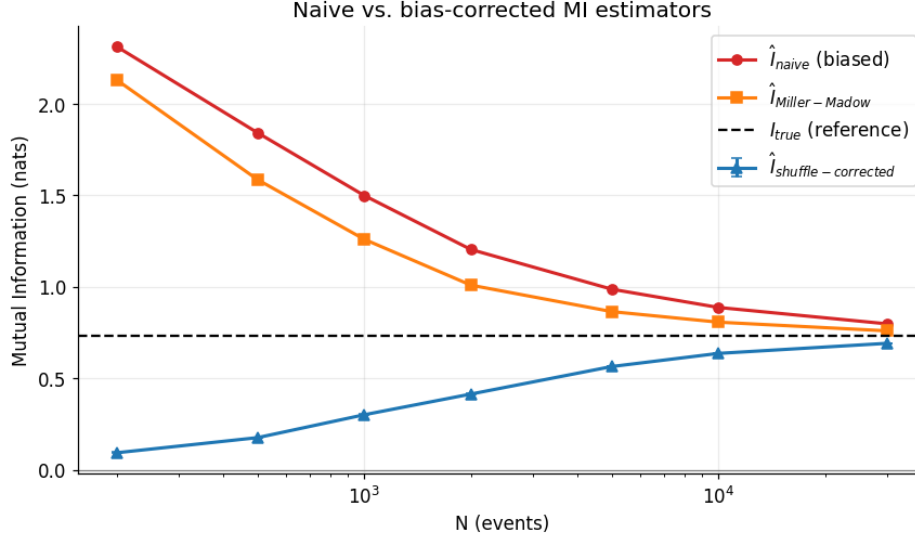


Figure 4: Naive vs. bias-corrected MI estimators across sample sizes.

6 The rigorous fix: NSB Bayesian entropy estimator

Both corrections above are still, fundamentally, **point corrections derived under a large- N assumption** – they estimate one number (the bias) and subtract it. The **Nemenman–Shafee–Bialek (NSB)** estimator [13] takes a genuinely different approach: instead of correcting a biased point estimate, it computes the **Bayesian posterior mean entropy** directly, using a prior specifically constructed to avoid smuggling in any implicit assumption about how spread-out the distribution is. This is the standard tool for the severely-undersampled regime ($N \lesssim K$) in the neuroscience spike-train MI literature – precisely our regime once hadron multiplicities are binned in 2D.

The construction, in four steps:

1. Put a symmetric Dirichlet prior $P(p|\beta) \propto \prod_i p_i^{\beta-1}$ over the K possible bin probabilities. For

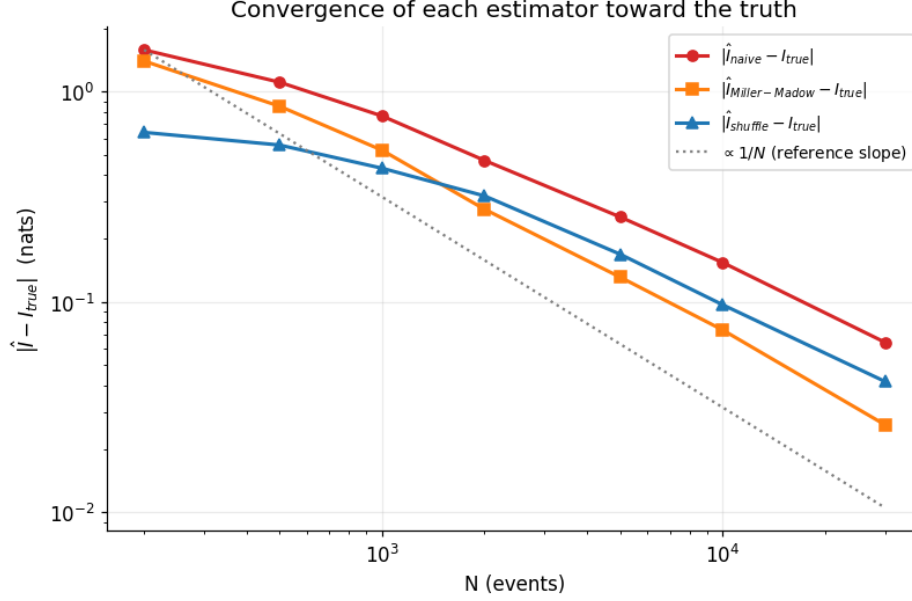


Figure 5: Convergence of each estimator toward the true value, on a log-log scale, with a $1/N$ reference slope.

any fixed β , the posterior mean entropy given counts $\{n_i\}$ has a closed form,

$$\langle S \rangle_{\beta, n} = \psi_0(N + K\beta + 1) - \sum_i \frac{n_i + \beta}{N + K\beta} \psi_0(n_i + \beta + 1). \quad (14)$$

2. **The catch:** a flat prior on β does *not* give a flat prior on entropy – for large K , $P(S | \beta)$ is sharply peaked as a function of β . Fixing any single β (as ordinary add-one/Laplace/Jeffreys smoothing does) silently assumes an entropy value before looking at the data.
3. NSB’s fix: reweight β by the hyperprior that makes the *induced* prior on S flat – literally the rate of change of the prior mean entropy with β ,

$$\rho(\beta) = K \psi_1(K\beta + 1) - \psi_1(\beta + 1). \quad (15)$$

4. Average over β , weighted by both this hyperprior and the Dirichlet–multinomial evidence $P(n | \beta)$,

$$\hat{S}_{\text{NSB}} = \frac{\int_0^\infty d\beta \rho(\beta) P(n | \beta) \langle S \rangle_{\beta, n}}{\int_0^\infty d\beta \rho(\beta) P(n | \beta)}. \quad (16)$$

This reduces to a tractable 1D numerical integral. We apply it separately to S_A, S_B, S_{AB} (each needs its own alphabet size K : for the joint distribution, $K_{AB}^{\text{alphabet}} = K_A^{\text{alphabet}} \times K_B^{\text{alphabet}}$, the full product space of possible outcomes – not the observed K_{AB} count from before, since NSB needs to know the size of the space being integrated over, not how much of it happened to be hit), then combine: $\hat{I}_{\text{NSB}} = \hat{S}_A^{\text{NSB}} + \hat{S}_B^{\text{NSB}} - \hat{S}_{AB}^{\text{NSB}}$.

NSB tracks I_{true} closely across the entire range, including at $N = 200$ where the naive estimator is off by roughly a factor of three and Miller–Madow has barely moved. This is the payoff of treating the whole problem as Bayesian inference over entropy rather than patching a biased point estimate

Table 3: All four MI estimators, head to head.

N	$\hat{I}_{\text{naive}}$	$\hat{I}_{MM}$	$\hat{I}_{\text{shuffle}}$	$\hat{I}_{\text{NSB}}$	I_{true} (ref.)
200	2.3511	2.1711	0.0578	0.5098	0.7329
500	1.7525	1.5035	0.2137	0.6833	0.7329
1000	1.5071	1.2711	0.3062	0.7369	0.7329
2000	1.2299	1.0367	0.4340	0.7416	0.7329
5000	1.0030	0.8842	0.5807	0.7678	0.7329
20000	0.8222	0.7716	0.6752	0.7411	0.7329

after the fact. Figure 6 adds NSB to the convergence comparison: its residual curve sits visibly below the other three methods across most of the range, and in particular stays far more stable at small N – it doesn’t need $N \gg K$ to behave sensibly.

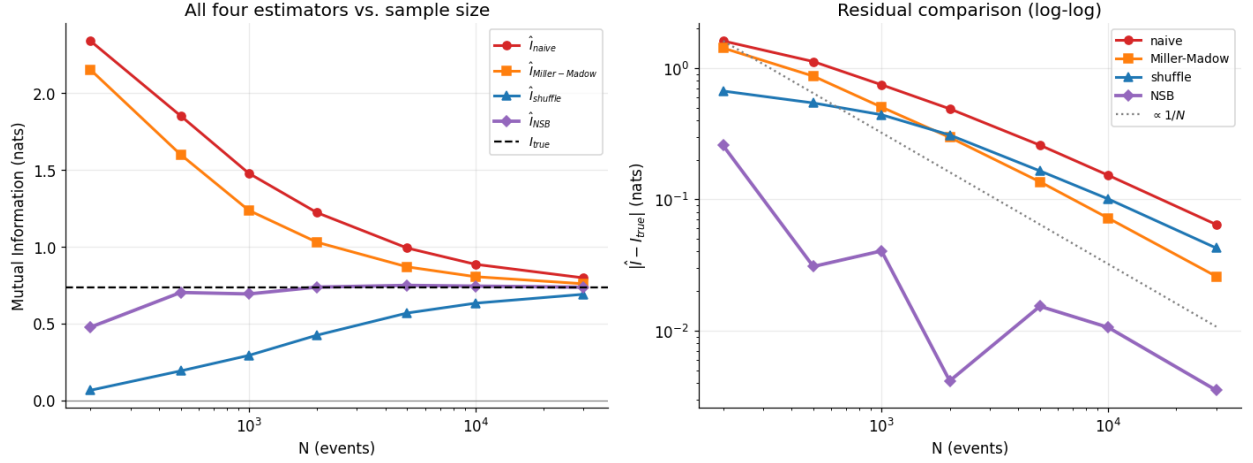


Figure 6: Left: all four estimators vs. sample size. Right: residual comparison on a log-log scale, with a $1/N$ reference slope.

Practical notes for the real STAR dijet data.

- The alphabet size K (for each marginal, and $K_A \times K_B$ for the joint) needs to be fixed *before* looking at the data – pick a generous fixed upper bound on plausible hadron multiplicity per jet (e.g. from detector acceptance / kinematic limits), not the observed range, or a data-dependent bias is reintroduced through the back door.
- NSB is somewhat more expensive per call than MM or shuffle-subtraction (a 1D numerical integral instead of a closed-form correction or a handful of permutations), but it remains a fraction of a second per event sample – entirely practical for a real analysis, including a full jackknife or bootstrap for uncertainty quantification.
- The same estimator applies directly to $S(B|A) = S_{AB} - S_A$ and to NMI – swap in the NSB entropies there instead of the naive/MM/shuffle versions before quoting any central value of these derived quantities.

7 Summary and recommendations from the toy Monte Carlo study

1. **The observables are classically bounded, and the bounds are not testable physics.** $S(B|A) \geq 0$ and $\text{NMI} \leq 1$ hold identically for any measured multiplicity distribution (Sec. 2.1); they cannot be used as quantum witnesses. The physically meaningful measurements are the central values themselves – $I(A:B)$, NMI, $S(B|A)$ – compared across kinematics, against generator baselines, and against entanglement-ansatz predictions (Sec. 1); proximity to the perfect-correlation limit $\text{NMI} \rightarrow 1$ is the Schmidt-limit signature, not a bound violation. Genuine quantum discord likewise cannot be computed from hadron-count histograms – any classical joint distribution gives discord $\equiv 0$ (Sec. 2).
2. **The naive plug-in MI estimator is always biased upward**, with leading term $\approx (K_{AB} - K_A - K_B + 1)/(2N)$. Finer binning or fewer events both make it worse – comparing different STAR running periods, luminosities, or binning choices will report different central MI values from what should be the same physics, purely from this effect.
3. **Miller–Madow** (done correctly, using the observed K_{AB} , not the product $K_A K_B$) and **shuffle-subtraction** both substantially reduce this bias and stay well-behaved, but neither is exact at small N : Miller–Madow is a first-order truncation; shuffle-subtraction tends to systematically overcorrect when the true correlation is strong.
4. **NSB** is the recommended estimator for the real analysis: a Bayesian posterior-mean entropy built on a prior specifically constructed to be uninformative about entropy itself. It tracks the truth closely across the entire tested N range, including deep in the severely-undersampled regime where the other methods struggle, at a modest computational cost.
5. **Before quoting any measured correlation** – especially one to be compared across datasets or against model predictions – bias-correct with NSB and validate the quoted uncertainty against the estimator’s own sampling noise floor at that N , obtained by bootstrap/jackknife on the dataset itself and calibrated by a closure test at the analysis’s actual statistics (Sec. 9).
6. **Any estimated $\widehat{S(B|A)} < 0$ or $\widehat{\text{NMI}} > 1$, at any apparent significance, is a finite-sample artifact by theorem** (Sec. 2.1). Its appearance is a diagnostic of residual estimator bias or underestimated uncertainty, not of physics; the correct response is more statistics or better correction, not interpretation.

8 Validation on PYTHIA 8 Monte Carlo simulation

The toy model of Secs. 4–6 demonstrates that the estimator pipeline behaves correctly on data with a known, simple generative structure. The natural next step is to check the same pipeline on a full QCD event generator, where hadron multiplicities arise from parton showering, hadronization, and (optionally) the underlying event – none of which was built into the toy model by hand. We generated back-to-back charged-jet dijets with PYTHIA 8 (Monash 2013 tune, anti- k_T jets, $R = 0.4$, charged final-state particles only, matching a STAR-TPC-like charged-jet definition) and applied the identical naive/ Miller–Madow/shuffle/NSB pipeline, unmodified, to the generator output.

This particular validation run used LHC-like kinematics ($\sqrt{s} = 14$ TeV rather than the $\sqrt{s} = 200$ GeV RHIC energy that motivates the eventual STAR measurement) specifically *because* of its much wider accessible jet- p_T range: as discussed below (Sec. 8.2), the multiplicity-vs- p_T correlation predicted by QCD is intrinsically a slow, logarithmic effect, and a wide dynamic range makes it far

easier to confirm the pipeline is tracking real physics rather than an artifact of a narrow kinematic window. The 200 GeV run for the actual STAR measurement is a separate, ongoing exercise (see Sec. 10); everything in this section is a *methods validation*, not a physics result.

8.1 Multiplicity and p_T distributions

Figure 7 shows the charged-multiplicity distributions for the leading and subleading jets, overlaid with a method-of-moments Negative-Binomial fit (as in Sec. 4). Both jets show genuine overdispersion relative to Poisson (var/mean = 1.55 for the leading jet, 1.29 for the subleading jet), qualitatively consistent with – though not identical in shape to – the NBD-like multiplicity distributions assumed in the toy model. This is a useful cross-check in its own right: the toy model’s central structural assumption (NBD-shaped marginals) is not merely a convenient fiction, but a reasonable approximation to what a full QCD generator actually produces.

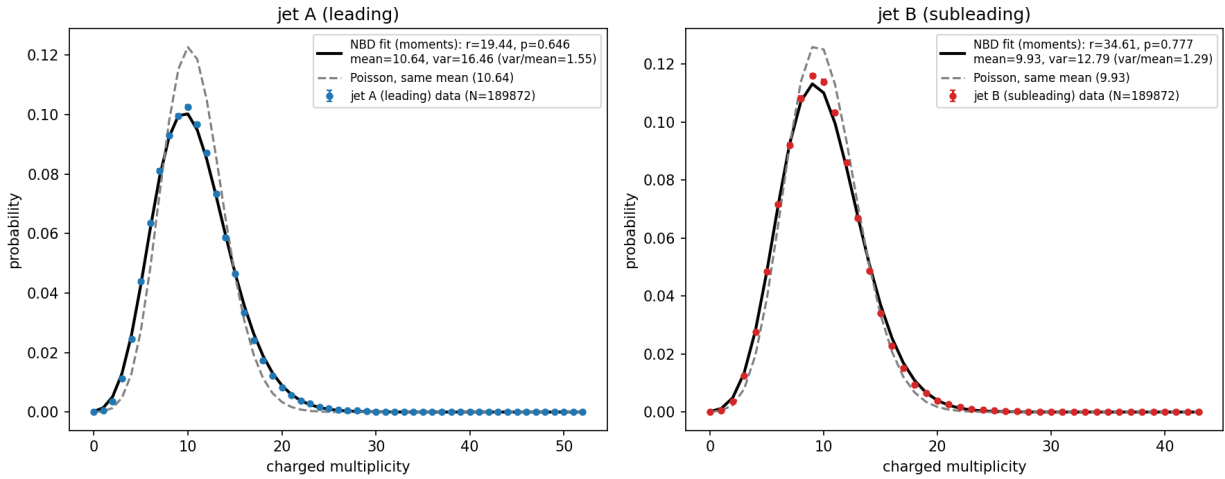


Figure 7: PYTHIA 8: charged-multiplicity distributions for the leading (left) and subleading (right) jet, with NBD and same-mean Poisson overlay (cf. the toy-model version, Sec. 4).

Figure 8 shows the corresponding joint distribution: 36.9% of the 53×44 grid is occupied ($K_{AB} = 861$) at $N \approx 1.9 \times 10^5$ events – a moderately, not severely, undersampled regime, comparable to the middle of the range explored in the toy study. Figure 9 shows the (steeply falling, as expected for a QCD jet spectrum) leading- and subleading-jet p_T distributions, and Fig. 10 their joint distribution – the latter makes visually clear why quantile (rather than fixed-width) p_T binning is necessary for the differential measurements below: the bulk of events sit in a narrow low- p_T band, with a long, sparse tail.

8.2 The multiplicity– p_T correlation

A basic physical expectation – confirmed experimentally (e.g. ALICE [14]: “the mean charged-particle multiplicity inside the leading jet cone rises monotonically with increasing jet p_T ”) – is that higher- p_T jets should carry higher charged multiplicity, via additional phase space for parton radiation before hadronization. This growth is predicted by QCD coherent-branching (MLLA) arguments [16] to be slow and roughly *logarithmic* in p_T , not linear, which matters directly for how easy the effect is to see over a limited kinematic range (as it was, initially, in an earlier 200 GeV validation attempt with a narrower accessible p_T window).

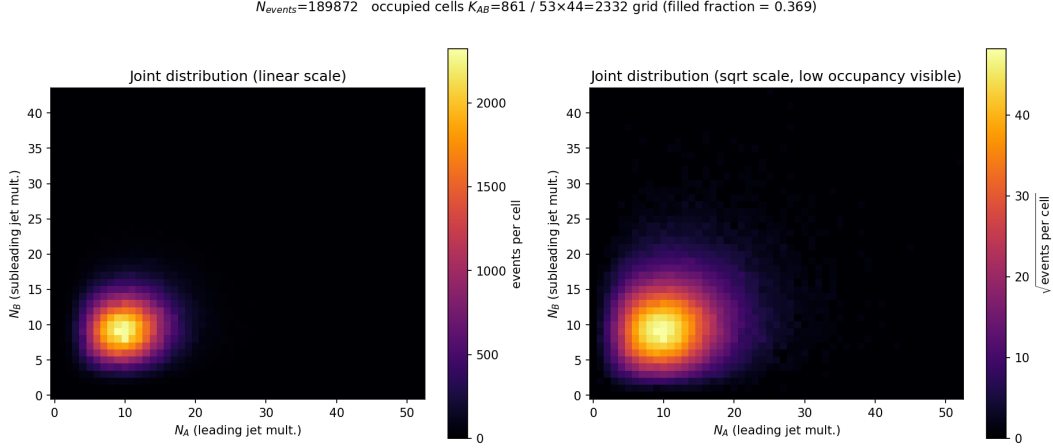


Figure 8: PYTHIA 8: joint charged-multiplicity distribution, linear and sqrt color scale (cf. Fig. 1).

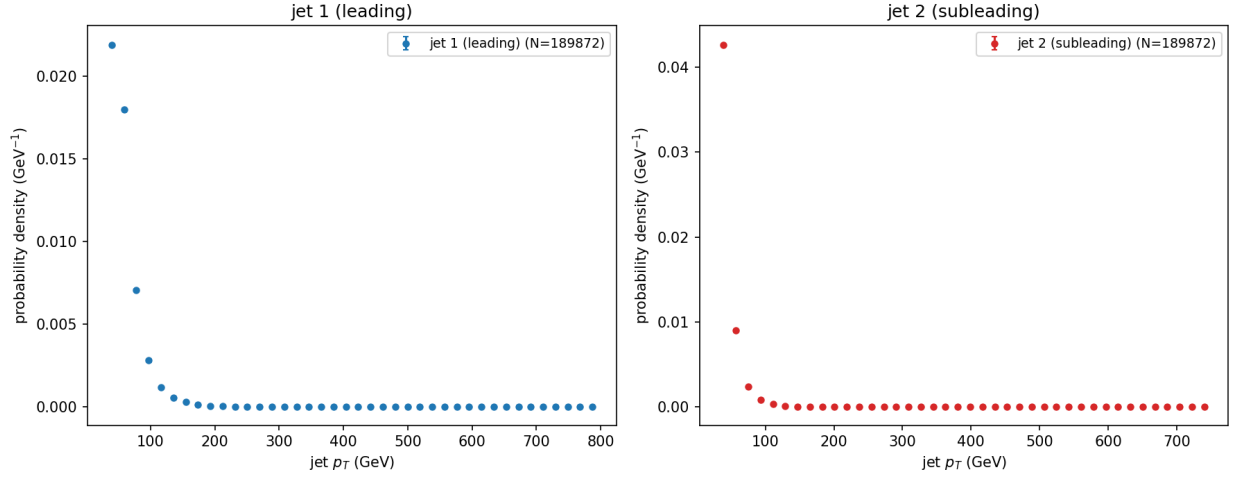


Figure 9: PYTHIA 8: leading- and subleading-jet p_T distributions.

Figure 11 shows the raw, unbinned-by-any-entropy- machinery profile of mean charged multiplicity vs. jet p_T , together with a weighted fit of the MLLA-motivated form $\langle N \rangle = a + b \ln p_T$. The fitted slope is highly significant for both jets ($b = 3.77 \pm 0.05$, 83σ , leading jet; $b = 4.71 \pm 0.07$, 66σ , subleading jet) – direct, quantitative confirmation that PYTHIA 8 reproduces the expected QCD multiplicity-growth pattern, and that the analysis pipeline correctly recovers it from generator output with no prior assumption beyond bin means and standard errors.

8.3 Finite-sample bias, reproduced on generator output

Figure 12 repeats the toy study’s core diagnostic (Sec. 4) directly on subsamples of the real PYTHIA 8 output: the naive $\hat{I}$ falls from ≈ 0.66 nats at $N = 200$ down toward the full-sample NSB value (≈ 0.009 nats) as N grows to $\approx 1.9 \times 10^5$ – the same qualitative bias curve seen throughout Sec. 4, now confirmed on a dataset the toy model was never built to resemble. This is a meaningful cross-check: the bias mechanism identified analytically (Sec. 3) and demonstrated on a synthetic Gamma-Poisson model is not an artifact of that specific generative structure – it appears with the

N_events = 189872

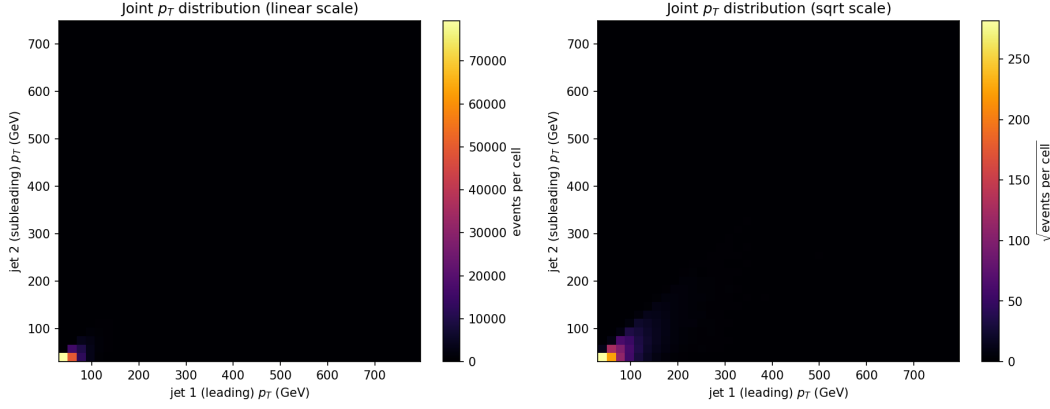


Figure 10: PYTHIA 8: joint leading/subleading jet p_T distribution, linear and sqrt color scale.

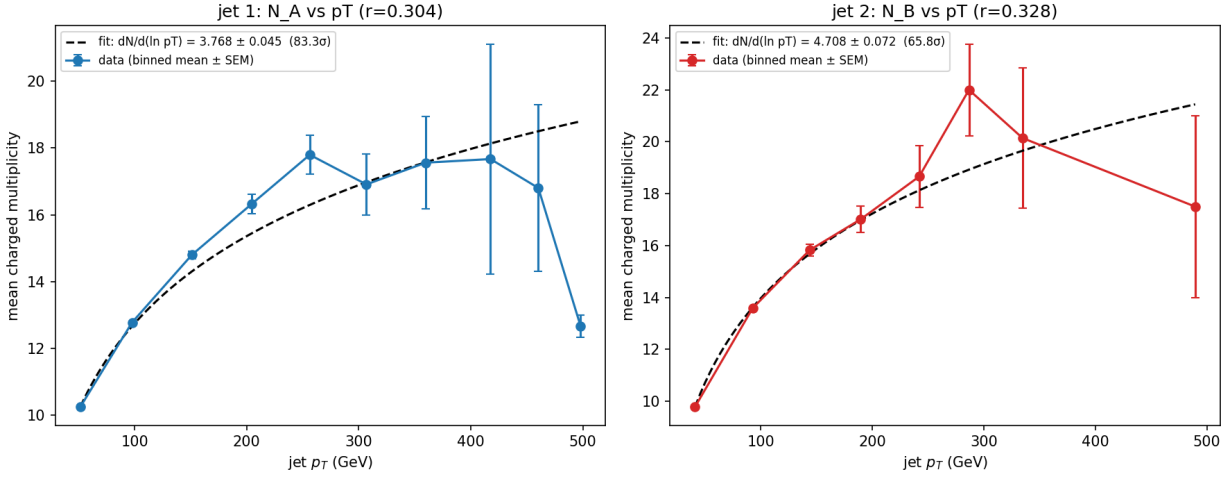


Figure 11: PYTHIA 8: raw mean charged multiplicity vs. jet p_T (binned mean $\pm$ SEM), with a weighted fit to $\langle N \rangle = a + b \ln p_T$. Both slopes are significant at $> 60\sigma$.

same shape on genuine QCD Monte Carlo output.

8.4 Point estimates on the full sample

Figure 13 shows all four estimators on the full sample ($N \approx 1.9 \times 10^5$): naive (0.0112 ± 0.0004), Miller–Madow (0.0092 ± 0.0004), shuffle-corrected (0.0084 ± 0.0003), and NSB (0.0085 ± 0.0004 nats) – naive sits visibly above the other three, as expected from the bias mechanism above, while Miller–Madow/shuffle/NSB agree closely with each other, consistent with all three having converged toward the true value at this statistics (matching the closure seen among independently-biased estimators at large N in the toy study, Sec. 4). The conditional entropy $S(B|A) = 2.65$ nats and $\text{NMI} \ll 1$ sit far from the perfect-correlation (Schmidt-limit) boundary: PYTHIA 8’s cross-jet multiplicity correlation is weak on the NMI scale. Since PYTHIA’s hadronization contains no entanglement by construction, these values constitute the classical QCD generator baseline of Sec. 1 against which the same measurement on real data can eventually be compared. This measurement

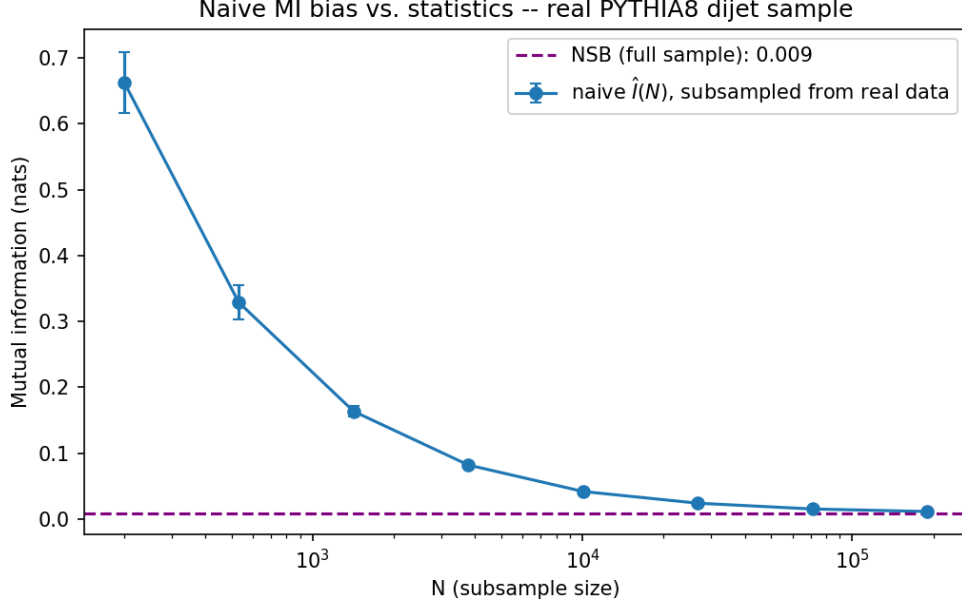


Figure 12: PYTHIA 8: naive $\hat{I}$ vs. subsample size N , real generator output (cf. the toy-model version, Fig. 2).

is a validation of the estimator machinery, not a physics claim.

8.5 Differential measurement vs. jet p_T

Figures 14–16 show S_A, S_B, S_{AB} , $I(A:B)$, and $S(B|A)$ (NSB) in quantile bins of leading-jet p_T . All three entropies rise smoothly with p_T (Fig. 14), directly reflecting the multiplicity growth of Sec. 8.2. $I(A:B)$ (Fig. 15) also rises with p_T , modestly at low-to-mid p_T and more sharply in the highest (widest, most statistics-limited) bin; $S(B|A)$ (Fig. 16) stays solidly positive and rising throughout, i.e. the correlation stays far from the perfect-correlation boundary at every p_T probed.

8.6 Multi-method comparison vs. p_T

Figure 17 overlays naive/Miller–Madow/ shuffle/NSB for $I(A:B)$, and naive/Miller–Madow/NSB for $S(B|A)$ (no shuffle analogue, Sec. 6), each with bootstrap error bars computed by the same unified procedure (Sec. 8.8). The bias hierarchy naive > Miller–Madow $\gtrsim$ NSB $\approx$ shuffle seen on the full sample (Fig. 13) persists across p_T bins, most visibly in the highest- p_T (lowest-statistics) bin, where naive’s upward bias is largest – exactly the pattern the toy study predicts for a bin with fewer events and a wider effective alphabet.

8.7 Cross-validation with an independent implementation, and normalized mutual information

The full pipeline was re-implemented independently in ROOT/C++ (with its own validated digamma/trigamma special functions, NSB posterior computation, and bootstrap machinery) as a cross-check against the Python implementation above, applied to the *same* generated event sample ($N \approx 1.9 \times 10^5$, identical to Secs. 8–8.2). This section reports the ROOT pipeline’s results for NMI (Sec. 2), which were not part of the original Python-only analysis above.

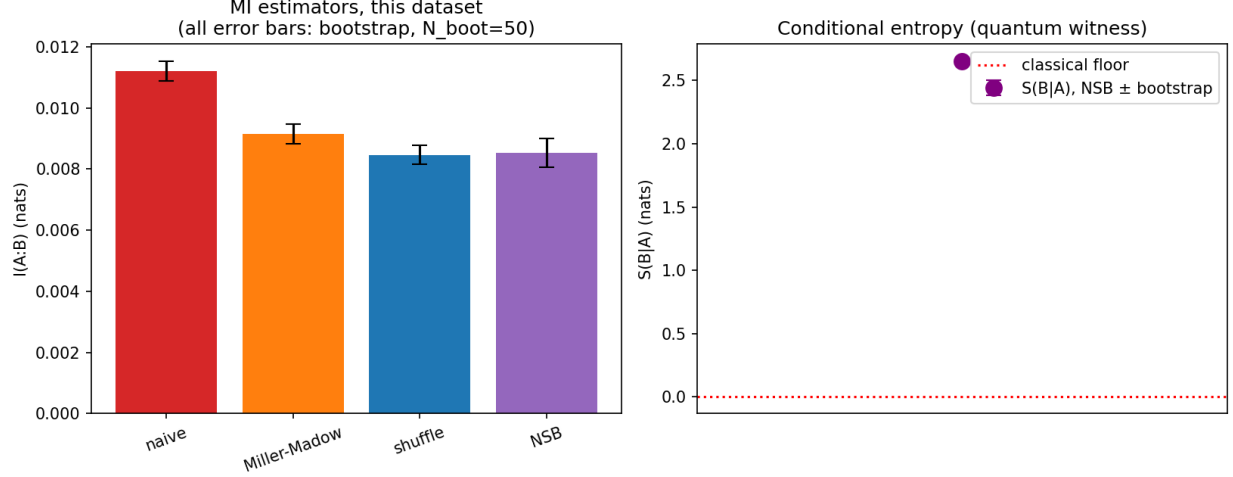


Figure 13: PYTHIA 8: all four MI estimators (left) and $S(B|A)$ (right) on the full sample, bootstrap error bars.

Figure 18 shows NMI (NSB) vs. leading-jet p_T : all values sit at $\text{NMI} \ll 1$, far below the perfect-correlation bound, consistent with the small $I(A:B)$ values already seen in absolute terms (Fig. 15) and, as before, expected for a classical event generator with weak cross-jet correlation. The value of NMI here is methodological rather than a physics result on this dataset: it demonstrates the normalized observable is measurable with the same NSB machinery as I and $S(B|A)$, and (Sec. 8.8 below) that doing so correctly requires some care.

8.8 A methodological pitfall: consistent uncertainty propagation for derived ratios

During development, the ROOT pipeline’s multi-method comparison plot for NMI (the NMI analogue of Fig. 17) showed NSB error bars roughly $5\text{--}6\times$ larger, relative to the central value, than NSB’s own error bars on $I(A:B)$ in the immediately adjacent panel (Fig. 19, left) – despite both quantities being built from the same S_A, S_B, S_{AB} on the same events. Tracing this down surfaced a genuine implementation bug, worth recording here since it is a general pitfall for any bootstrap-based uncertainty on a *derived, nonlinear* quantity (a ratio, in this case), not specific to this codebase.

The bug. $I(A:B)$ ’s bootstrap uncertainty was correctly computed by resampling the event sample N_{boot} times and recomputing I fresh on each resample. NMI’s point estimate used the same procedure, but its *uncertainty* was accidentally taken from an older, separate analytic error-propagation formula (independence approximation on $\text{Var}(I)$ and $\text{Var}(\min(S_A, S_B))$ combined in quadrature) left over from an earlier version of the pipeline – a different, more conservative methodology than the bootstrap used for I and $S(B|A)$ in the very same function, silently mixed into one comparison plot.

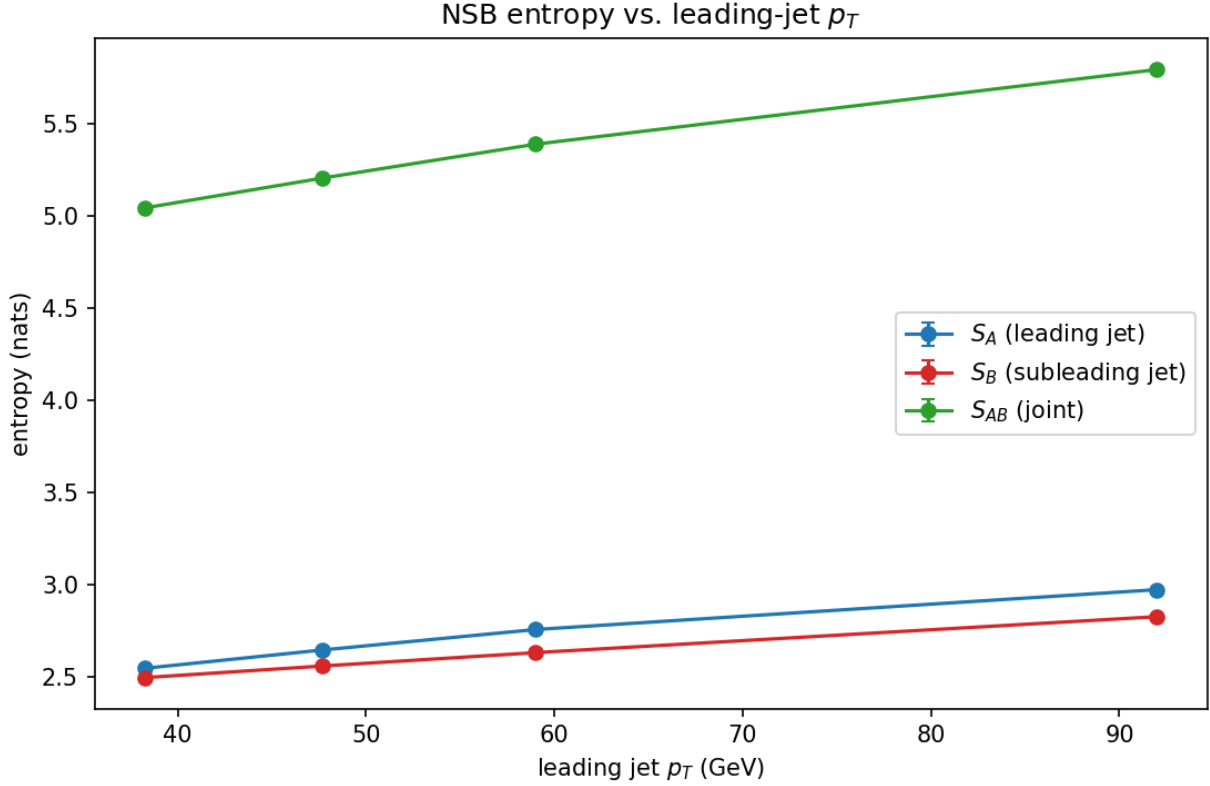


Figure 14: PYTHIA 8: S_A, S_B, S_{AB} (NSB) vs. leading-jet p_T .

The fix. NMI is now bootstrapped exactly like I and $S(B|A)$: each resample’s already-computed S_A, S_B (needed anyway for that resample’s I) are reused to form

$$\text{NMI}_{\text{resample}} = \frac{I_{\text{resample}}}{\min(S_{A,\text{resample}}, S_{B,\text{resample}})}, \quad (17)$$

and the spread of $\text{NMI}_{\text{resample}}$ across resamples gives its uncertainty – at no extra computational cost, since S_A, S_B were already being computed for I ’s own bootstrap. Figure 19 illustrates the discrepancy that led to identifying this; Figure 20 confirms the fix.

The general lesson. When an uncertainty is obtained by bootstrap resampling, every derived quantity built from the same underlying estimate should be bootstrapped *together, from the same resamples* – not by mixing bootstrap for some derived quantities and analytic error propagation for others, even when both individually seem reasonable. This also happens to be the more *correct* choice statistically, not merely the more consistent one: bootstrapping a ratio directly captures the true correlation between numerator and denominator (both estimated from the same event sample), whereas independence- approximation error propagation assumes that correlation away and is therefore both methodologically inconsistent *and* typically biased toward overly conservative (too large) uncertainties. This pitfall generalizes beyond NMI: any pipeline computing multiple derived statistics from one bootstrap procedure should verify that *all* of them are actually using the same resamples, not just that each individually looks plausible.

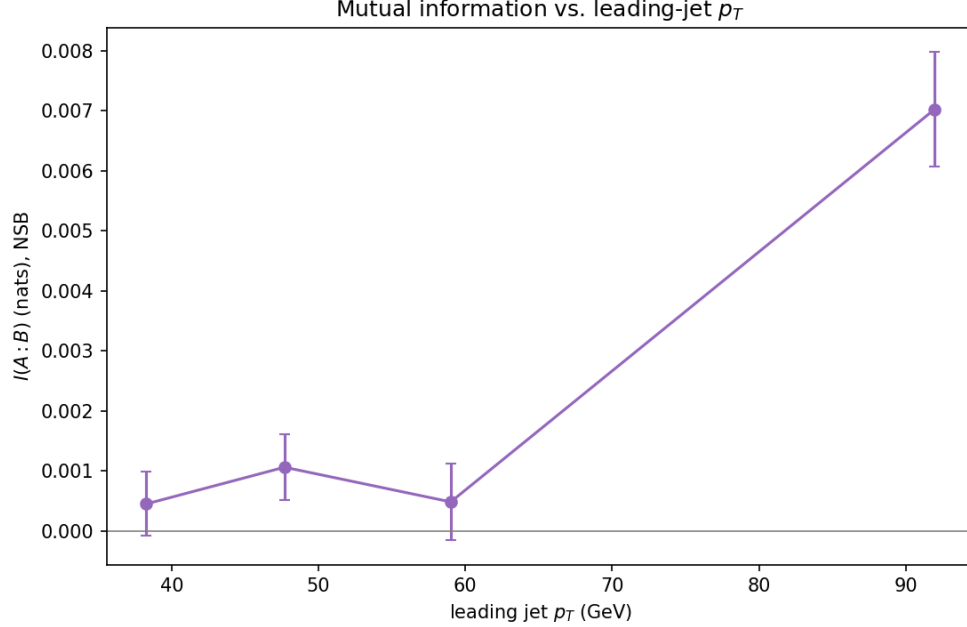


Figure 15: PYTHIA 8: $I(A:B)$ (NSB) vs. leading-jet p_T .

9 Tier-2 closure test: calibrating estimator uncertainty on realistic data

The toy study (Secs. 4–6) validates the estimator suite against a *known* true MI, by construction. Real PYTHIA 8 output, and eventually real STAR/ALICE data, has no such known truth to validate against directly. This section describes a closure test that closes that gap as far as is practically possible: calibrate a synthetic model *to resemble the real data*, use that model to generate a pseudo-truth precise enough to treat as exact, and check whether the estimator pipeline recovers it under closure – i.e. under exactly the statistical conditions (sample size, alphabet, correlation strength) of the real measurement.

9.1 Method

1. **Calibrate.** Fit the toy study’s Gamma-Poisson shared-latent model (Sec. 4: $\Lambda \sim \text{Gamma}(k, \theta)$, $N_A|\Lambda \sim \text{Poisson}(\Lambda)$, $N_B|\Lambda \sim \text{Poisson}(c\Lambda)$) to the *real* data’s five leading moments – $\langle N_A \rangle$, $\text{Var}(N_A)$, $\langle N_B \rangle$, $\text{Var}(N_B)$, $\text{Corr}(N_A, N_B)$ – via least squares over (k, θ, c) . This is a genuine, honest constraint: a 3-parameter model cannot exactly match 5 targets in general, and the achieved fit quality is reported explicitly rather than assumed.
2. **Establish a pseudo-truth.** Draw a very large sample (typically 10^6 – 5×10^6 events) from the *fitted* model. At this scale the naive estimator’s own bias ($\mathcal{O}(1/N)$, Sec. 3) is negligible, so $I, S(B|A), \text{NMI}$ computed on this sample are treated as numerically exact ($I_{\text{true}}, S(B|A)_{\text{true}}, \text{NMI}_{\text{true}}$) – calibrated to resemble the real measurement’s shape and correlation strength, not an arbitrary toy scenario.
3. **Closure.** Draw N_{repeats} independent fresh samples from the fitted model, each at the real dataset’s actual event count N (and using the real analysis’s alphabet), and run the full

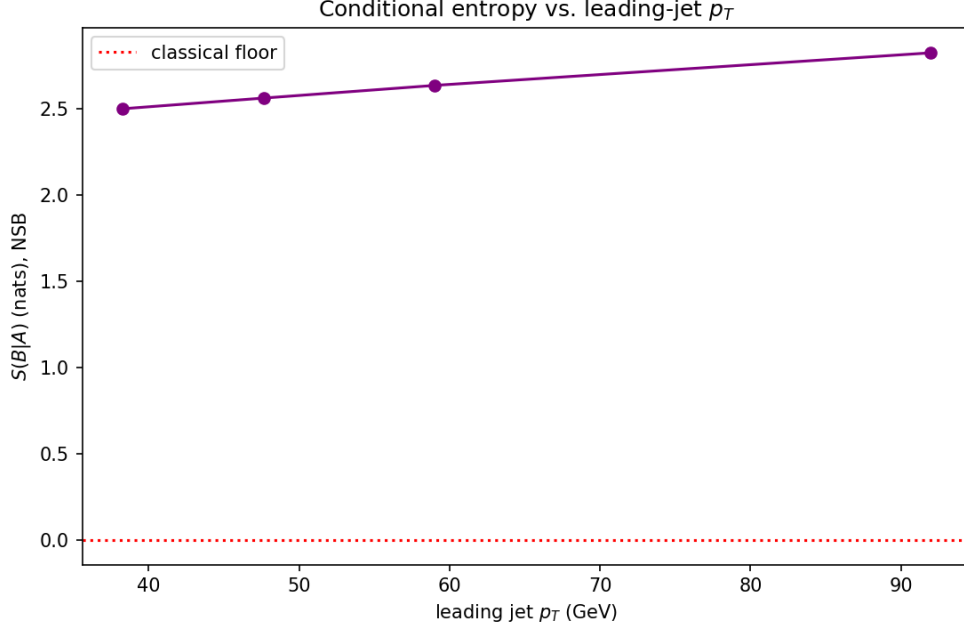


Figure 16: PYTHIA 8: $S(B|A)$ (NSB) vs. leading-jet p_T .

naive/Miller–Madow/shuffle/NSB pipeline on each.

4. **Check bias and calibration.** For NSB specifically, form the *pull* for each repeat, $z = (\hat{X}_{\text{NSB}} - X_{\text{true}})/\sigma_{\text{NSB}}$ for $X \in \{I, S(B|A), \text{NMI}\}$, using that repeat’s own bootstrap σ_{NSB} (Sec. 8.8). If the estimator is unbiased and its uncertainty well-calibrated, the pull distribution across repeats should be standard normal: mean 0, standard deviation 1, with $\sim 68\%$ of repeats within $\pm 1\sigma$ and $\sim 95\%$ within $\pm 2\sigma$.

This is purely diagnostic throughout – no bias or coverage correction is applied to any reported number anywhere in this note; the closure test’s role is to establish *whether* the raw bootstrap uncertainty can be trusted at face value for a given $(N, \text{alphabet})$, not to patch it.

9.2 Whole-sample results

Applying this to the full 14 TeV validation sample (Fig. 21, $I(A:B)$; Fig. 22, NMI) gives two consistent findings:

- **A mean pull of approximately -1 :** NSB systematically under-estimates $I(A:B)$ (and correspondingly NMI) by roughly one of its own reported bootstrap- σ ’s, on average, at this sample’s statistics. This is a real, non-negligible bias, though modest in scale.
- **A pull standard deviation of approximately 1.5–1.7:** the *actual* spread of $(\hat{X} - X_{\text{true}})$ across independent repeats is 50–70% *larger* than what the reported bootstrap σ claims. In other words, $\text{NSB} \pm \text{bootstrap } \sigma$ under-covers – the quoted error bars are too tight. Any uncertainty quoted from the raw bootstrap σ at this $(N, \text{alphabet})$ therefore understates the true uncertainty by roughly this same factor (e.g. a nominal 3σ separation between two measurements would correspond to a true separation closer to $3/1.7 \approx 1.8\sigma$).

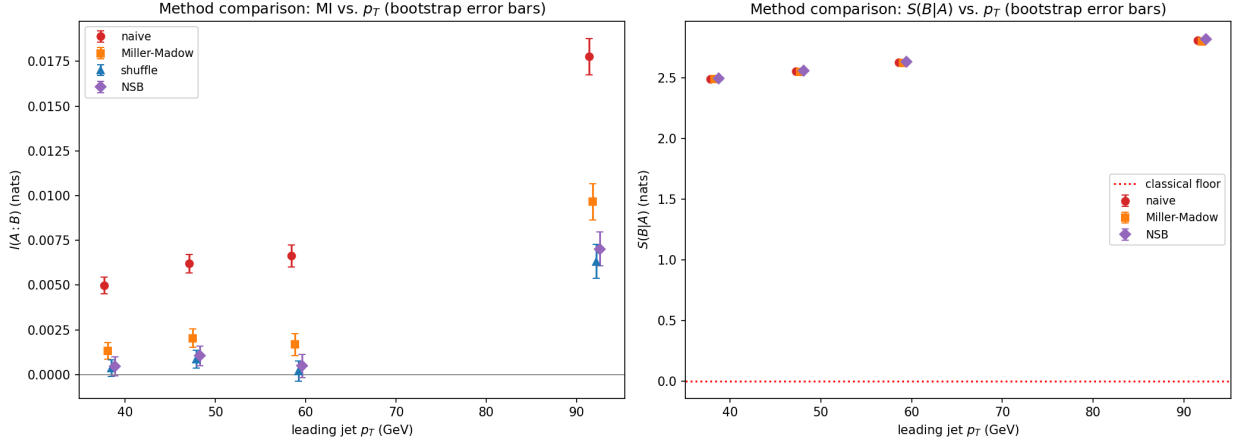


Figure 17: PYTHIA 8: method comparison for $I(A:B)$ (left) and $S(B|A)$ (right) vs. leading-jet p_T , bootstrap error bars throughout.

The pull distributions in both figures are also visibly non-Gaussian: skewed, with a heavier negative tail extending to several σ beyond what a standard normal would predict, rather than a uniformly-widened but still-symmetric Gaussian. This matters practically: a single multiplicative correction to σ (calibrating only the second moment) would not fully correct the shape of the tail, which is the part most relevant to any strong claim of a difference between datasets or between data and a model baseline. Any downstream use of these bootstrap uncertainties should treat this as a known, quantified limitation rather than an assumption to be taken on faith – exactly the purpose of running this closure test before, not after, a real measurement.

9.3 Extension to the conditional entropy directly

Sections 9.1–9.2 were run for $I(A:B)$ and NMI; since $S(B|A)$ is the derived quantity whose central value will be most directly interpreted physically – and, per Sec. 2.1, the one where residual estimator bias can masquerade as spurious boundary-approaching structure – a dedicated closure test for $S(B|A)$ itself (mean pull, pull standard deviation, and calibration fraction within $\pm 1\sigma/\pm 2\sigma$, identical procedure to Sec. 9.1) has been implemented and is not inferred from I ’s behavior via $S(B|A) = S_B - I(A:B)$ – that inference would carry S_B ’s own separate estimation noise and is not a substitute for a direct check. Numerical results for $S(B|A)$ ’s pull distribution on this sample will be reported in a future revision of this note.

9.4 p_T -differential closure test

The results of Sec. 9.2 calibrate NSB’s uncertainty for a dataset shaped like the *whole* sample – but both the underlying multiplicity/correlation structure and the per-bin event count change with p_T (Secs. 8.2, 8–8.7), so whole-sample calibration does not automatically transfer to any individual p_T bin, particularly the highest- p_T , lowest-statistics bins where the correlation structure is most interesting and least well-measured. A p_T -differential extension of the closure test has been implemented: for each p_T bin independently, fit a bin-specific Gamma-Poisson model to that bin’s own events (not the whole-sample fit), generate a bin-specific pseudo-truth, and run the closure procedure at that bin’s actual event count – directly answering “is NSB’s uncertainty trustworthy in *this* bin” rather than assuming the whole-sample answer applies uniformly.

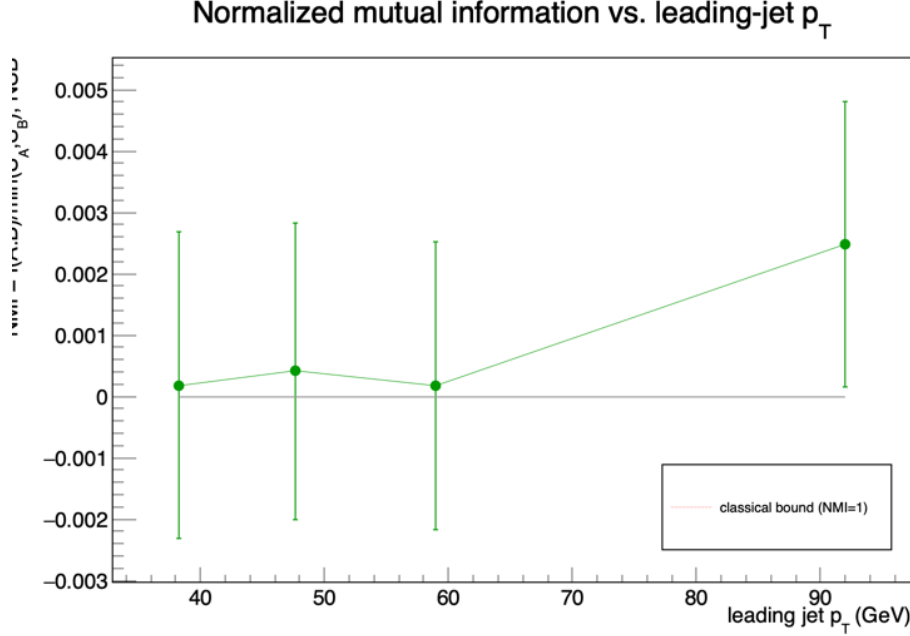


Figure 18: PYTHIA 8 (ROOT implementation): $NMI = I(A:B)/\min(S_A, S_B)$ (NSB) vs. leading-jet p_T . All values are far below the perfect-correlation bound $NMI = 1$, as expected for a classical event generator with weak cross-jet correlation.

Figure 23 shows the result, in the same four p_T bins used throughout Sec. 8.2 (p_T centers $\approx 38, 48, 59, 92$ GeV). The pattern is notably different from, and in one respect more encouraging than, the whole-sample result of Sec. 9.2:

- **Pull means** for $I(A:B)$ and NMI are small and negative throughout (≈ -0.12 to -0.17 in the three lower- p_T bins, shrinking to ≈ -0.04 in the highest- p_T bin) – markedly smaller in magnitude than the whole-sample ≈ -1 found in Sec. 9.2. $S(B|A)$ ’s pull mean is somewhat larger at low p_T (≈ -0.29) but trends toward zero as p_T increases, reaching ≈ 0 in the highest- p_T bin.
- **Pull standard deviations** for $I(A:B)$ and NMI rise from ≈ 0.58 (lowest- p_T bin, i.e. *over-covering* – the bootstrap σ is if anything too conservative there) through ≈ 0.80 and ≈ 0.95 to ≈ 0.92 in the highest- p_T bin – staying at or below 1 throughout, in contrast to the whole-sample 1.5–1.7 under-covering. $S(B|A)$ ’s pull standard deviation is close to 1 in every bin ($\approx 1.02, 1.00, 1.11, 0.92$) – the best-calibrated of the three quantities here, and encouragingly close to ideal given $S(B|A)$ is the headline derived quantity of the eventual measurement.

This bin-by-bin result being better-calibrated than the whole-sample one is itself a meaningful, self-consistent finding, not merely a nicer number: Sec. 9.4’s entire motivation was that the whole-sample Gamma-Poisson fit has to compromise across a range of p_T -dependent correlation structures at once (Sec. 8.2), whereas each bin-specific fit only has to describe its own, comparatively homogeneous sub-population. A better-calibrated per-bin result is exactly what that design difference predicts, and is evidence the whole-sample closure test’s miscalibration (Sec. 9.2) is at least partly a model-fit-quality effect specific to forcing one 3-parameter model to span the full p_T range, rather than a property of NSB’s bootstrap uncertainty in general. That said, this is a single

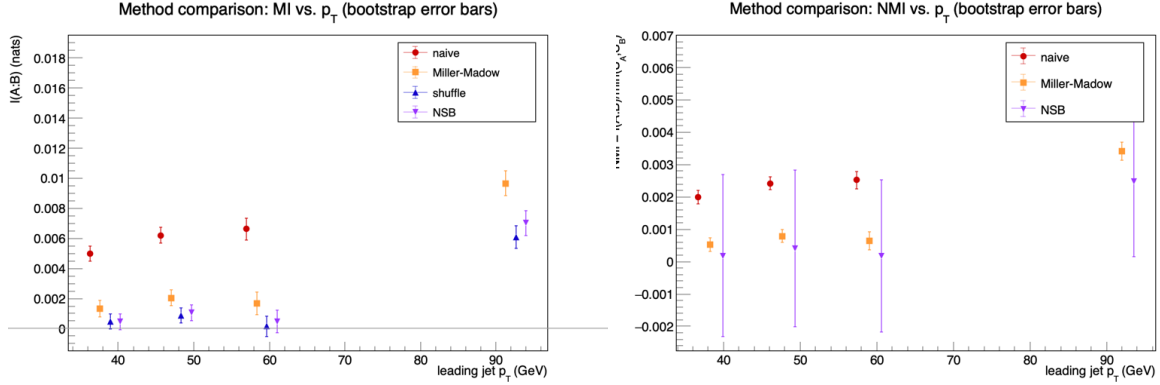


Figure 19: The bug, illustrated: $I(A:B)$ method comparison (left, unaffected) alongside the *pre-fix* NMI method comparison (right) from the same pipeline run. NSB’s NMI error bars are visibly far wider, relative to the central values, than I ’s – despite both being derived from the same underlying quantities on the same events. This discrepancy was the symptom that led to identifying the mixed-methodology bug described in the text.

validation dataset at one set of statistics; whether this pattern holds at STAR’s actual 200 GeV kinematics and statistics (Sec. 10) is not yet established and should not be assumed.

Even with this more encouraging per-bin picture, no bin here reaches pull std = 1 exactly, and the lowest- p_T bin’s over-covering for I/NMI (≈ 0.58) is itself a $\sim 40\%$ deviation from ideal calibration, in the conservative direction. The practical recommendation of Sec. 9.2 therefore still stands as the safe default – treat raw bootstrap σ with caution and consult a closure test at the actual bin/statistics in use – with this section’s result showing that the correction, at least for this validation sample, tends to be smaller and bidirectional per bin rather than a uniform large under-coverage.

10 Overall conclusions and next steps

The toy Monte Carlo study (Secs. 4–6) establishes the estimator methodology and its failure modes under controlled, known conditions. Section 8 shows the same pipeline, unmodified, correctly recovers physically-expected behavior (NBD-like overdispersion, MLLA-consistent multiplicity growth with p_T , the same finite-sample bias shape) on realistic QCD Monte-Carlo output where no such structure was hand-built in – a meaningful, if indirect, validation that the pipeline is measuring real structure rather than an artifact of the toy model’s specific construction. An independent ROOT/C++ re-implementation (Sec. 8.7) cross-validates the Python results and adds NMI, the scale-comparable normalized version of the observable (Sec. 2) needed to compare measurements across p_T bins, run periods, or collision energies on a common, bounded scale.

That cross-validation also surfaced a genuine implementation bug (Sec. 8.8): a bootstrap-based derived quantity (NMI) picked up its uncertainty from an inconsistent, more conservative analytic formula rather than the same resampling procedure used for $I(A:B)$ and $S(B|A)$ in the same function. The fix, and more importantly the general principle it illustrates – every derived quantity built from one bootstrap procedure must be resampled together, not mixed with a separately-derived analytic error – is recorded in full since it is a pitfall relevant to any bootstrap-based entropy/MI pipeline, not just this one.

Section 9 closes the gap the previous revision of this note flagged as open: a closure test

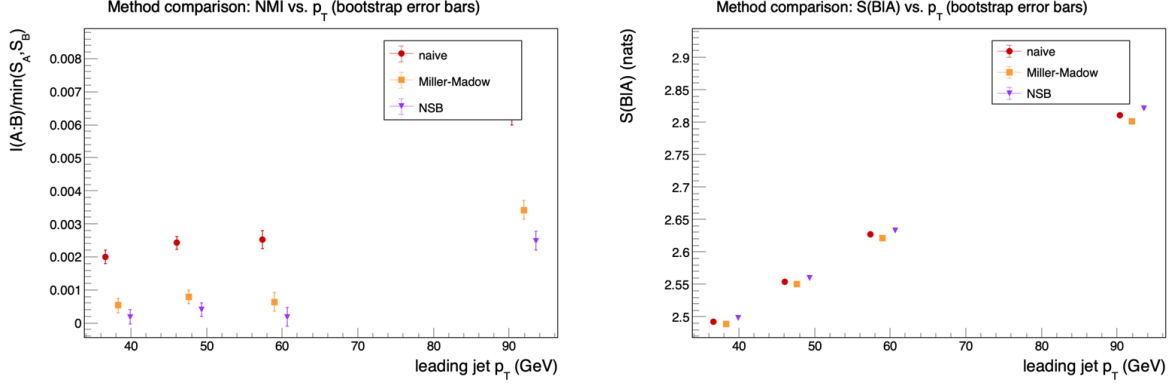


Figure 20: Post-fix confirmation, same pipeline and dataset as Fig. 19: NMI method comparison (left) now shows NSB error bars comparable in relative size to naive/Miller–Madow’s, in line with $I(A:B)$ ’s panel in Fig. 19 – the mixed-methodology discrepancy is gone. $S(B|A)$ method comparison (right, never affected by this particular bug) shown alongside for reference, confirming consistent bootstrap behavior across all three quantities.

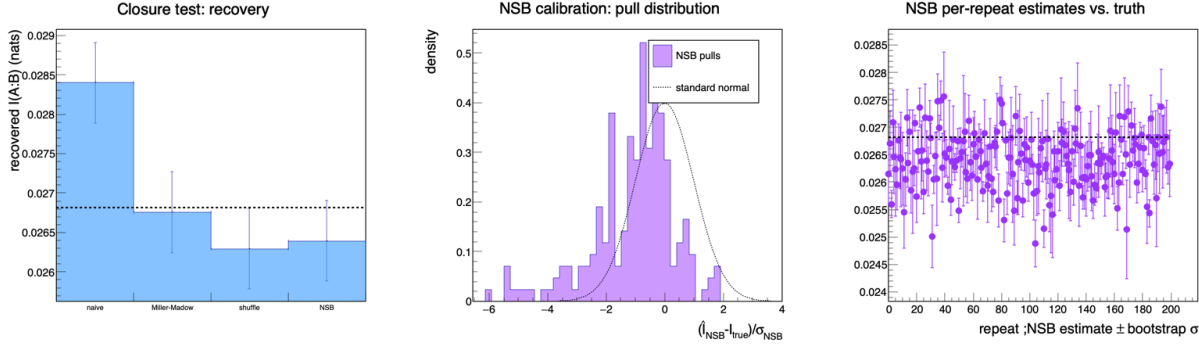


Figure 21: Whole-sample closure test, $I(A:B)$: recovery across methods (left), NSB pull distribution vs. standard normal (center), per-repeat NSB estimates vs. the fitted-model pseudo-truth (right).

calibrated to resemble the real PYTHIA 8 sample (Gamma-Poisson model fit to the real data’s marginals and correlation strength, huge reference sample for a pseudo-truth, repeated closure at the real dataset’s actual statistics) finds that NSB’s bootstrap uncertainty, at this sample’s $(N, \text{alphabet})$, is **under-covering by a factor of roughly 1.5–1.7**, with a mean bias of order $-\mathbf{1}\sigma$ and a pull distribution that is skewed rather than symmetrically widened. This is a quantitative, load-bearing result for any future comparison of measured correlations across datasets or against model baselines: a nominal 3σ difference computed from raw bootstrap σ at comparable statistics should be read as $\sim 1.8\text{--}2\sigma$ once this calibration factor is accounted for. Per Sec. 9, no correction has been baked into the pipeline itself – this factor is reported here as a measured property of the estimator at these statistics, to be applied explicitly and cited whenever an uncertainty is quoted, rather than silently folded into the numbers.

The p_T -differential extension of the closure test (Sec. 9.4) is now complete for this validation sample, and adds an important nuance: bin-by-bin, NSB’s pull standard deviations sit much closer to the ideal value of 1 ($\approx 0.58\text{--}0.95$ for $I(A:B)$ and NMI; $\approx 0.92\text{--}1.11$ for $S(B|A)$) than the whole-sample result above, consistent with each bin’s own Gamma-Poisson fit describing a more homogeneous sub-population than the one compromise fit spanning the whole p_T range has to.

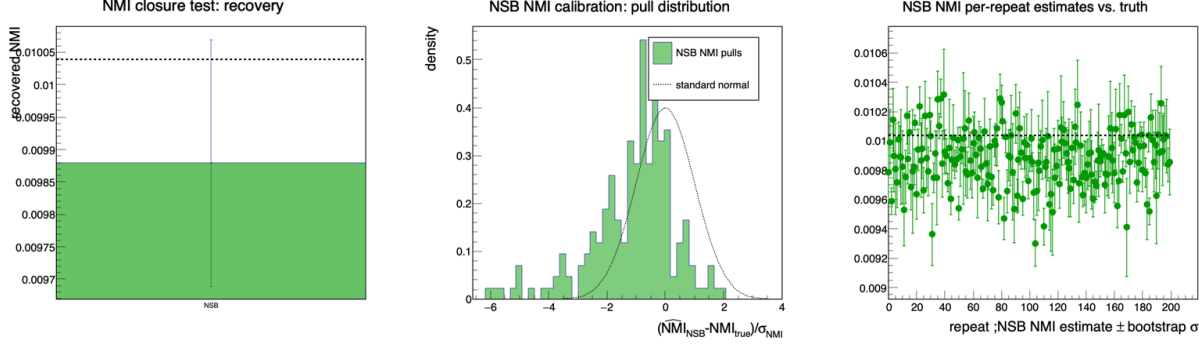


Figure 22: Whole-sample closure test, NMI: same three panels as Fig. 21, for the normalized observable.

This suggests the whole-sample under-coverage found in Sec. 9.2 is at least partly a consequence of fitting one model across a range of p_T -dependent structure, rather than an intrinsic property of NSB’s bootstrap procedure – but this is one validation dataset at one set of statistics, and the pattern should be re-checked, not assumed, at STAR’s actual kinematics. A dedicated whole-sample pull-distribution check for $S(B|A)$ itself (Sec. 9.3 – distinct from its per-bin counterpart just described, and still useful as the single most directly load-bearing calibration number for the headline quantity) remains to be run and will be reported in a future revision.

Remaining steps toward the actual STAR measurement: (i) repeat Sec. 8 at $\sqrt{s} = 200$ GeV with STAR-appropriate jet kinematics rather than the wide-dynamic-range 14 TeV validation kinematics used here; (ii) apply the detector-level acceptance and tracking-efficiency model (already implemented in the generator, not yet exercised in this note) and repeat the full analysis chain; (iii) rerun the whole-sample and p_T -differential closure tests (Secs. 9.2–9.4) at the actual statistics of (i)–(ii), obtaining calibration factors specific to the real analysis rather than relying on the 14 TeV validation sample’s factors as a proxy; (iv) once (i)–(iii) are complete, apply the identical pipeline to real STAR data, with the closure-test calibration factors from (iii) applied explicitly before unblinding, so that every quoted $I(A:B)$, NMI, and $S(B|A)$ value carries a defensible, calibration-corrected uncertainty when compared against generator baselines and the entanglement-ansatz expectations of Sec. 1.

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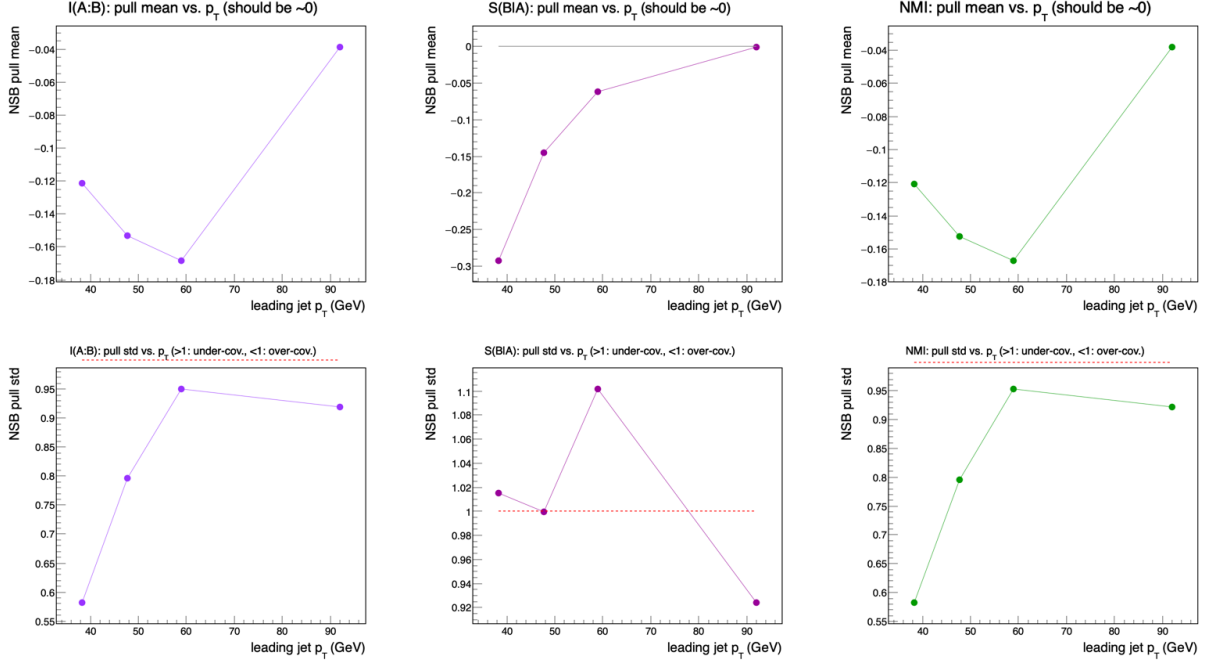


Figure 23: p_T -differential closure test: NSB pull mean (top row) and standard deviation (bottom row) vs. leading-jet p_T , for $I(A:B)$ (left), $S(B|A)$ (center), and NMI (right). Red dashed line marks perfect calibration (pull std = 1).

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